

KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM

(An Autonomous Institution)

Approved by AICTE, Affiliated to Anna University, Chennai.

Accredited by NBA (CSE, ECE, EEE & MECH), Accredited by NAAC with „A“ Grade

KIOT Campus, Kakapalayam – 637 504. Salem Dt., Tamil Nadu, India.



M.E. / M.Tech. Regulations 2023

M.E. – Industrial Safety Engineering

Curriculum and Syllabi

(For the Students Admitted from the Academic Year 2023 – 2024 onwards)

Version: 2.0	Date: 15.12.2025
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M.E. / M.Tech. / M.C.A. REGULATIONS 2023 (R 2023)
CHOICE BASED CREDIT SYSTEM AND OUTCOME BASED EDUCATION

M.E.- INDUSTRIAL SAFETY ENGINEERING

VISION OF THE INSTITUTE

- To be a world-class institution to impart value and need based professional education to the aspiring youth and carving them into disciplined world class professional who have the quest for excellence, achievement orientation and social responsibilities.

MISSION OF THE INSTITUTE

A	To promote academic growth by offering state-of-art undergraduate, postgraduate and doctoral programs and to generate new knowledge by engaging in cutting – edge research.
B	To nurture talent, Innovation, entrepreneurship, all-round personality and value system among the students and to foster competitiveness among students.
C	To undertake collaborative projects which offer opportunities for long-term interaction with academia and industry for creating a sustainable world.
D	To pursue global standards of excellence in all our endeavors namely teaching, research, consultancy, continuing education and support functions.

VISION OF THE DEPARTMENT

To create competent and industry relevant Mechanical Engineers with professional and social values to meet global challenges.

MISSION OF THE DEPARTMENT

M1	Enabling environment for effective teaching - learning and research to meet global challenges.
M2	Motivating students to pursue higher education and to excel in competitive examinations and entrepreneurship.
M3	Establish a continuous Industry Institute Interaction to make the students employable.
M4	Inculcate the student's leadership quality with ethical values and spirit of team work.


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PROGRAM EDUCATIONAL OBJECTIVES (PEOs)	
PEO 1	Possess a mastery of Health safety and environment awareness and safety management skills, to reach higher levels in their profession.
PEO 2	Proficient safety Engineer rendering professional expertise to the industrial and societal needs at national and global level subject to legal requirements.
PEO 3	Well communicate the information on Health safety and environment facilitating collaboration with experts across various disciplines so as to create and execute safe methodology in complex engineering activities.
PEO 4	Demonstrate professional and ethical attitude with awareness of current legal issues by rendering expertise to wide range of industries.

PROGRAM OUTCOMES (POs)	
Engineering Graduates will be able to:	
PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	Create, select, learn and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling to safety, health and environmental engineering activities with an understanding of the limitations.
PO 5	Demonstrate the societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to occupational health and safety practices.
PO 6	Recognize the need for, and have the preparation and ability to engage in life-long learning independently, with a high level of enthusiasm and commitment to improve knowledge and competence continuously.


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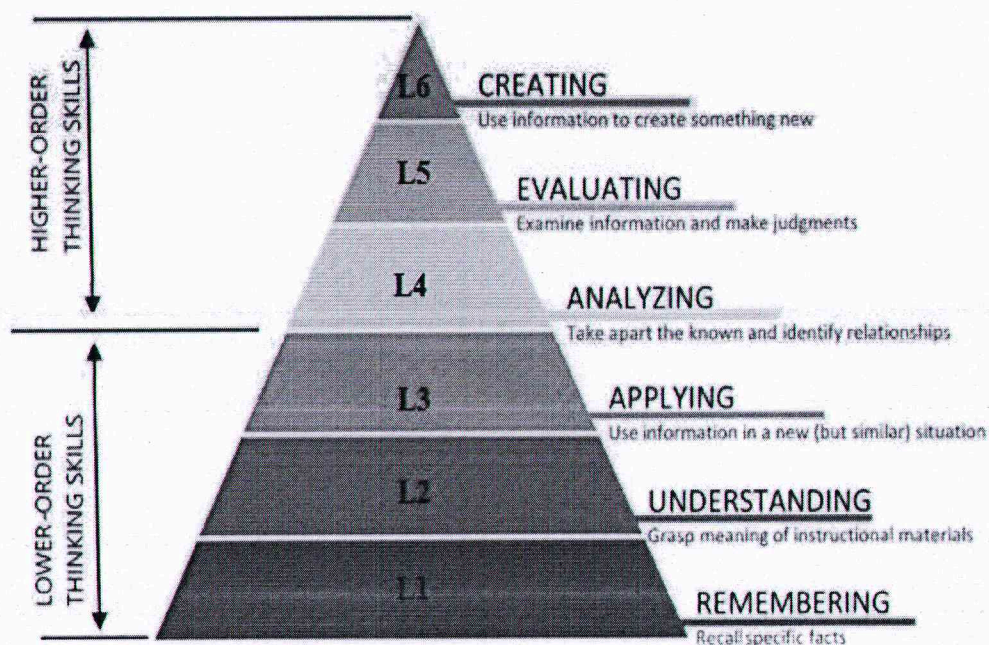
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BLOOM'S TAXONOMY LEVELS AND ACTION VERBS

(A) BLOOM'S TAXONOMY LEVELS (BTL):

Bloom's Taxonomy (BT) is based on the belief that learners must first acquire basic foundational knowledge about a subject before progressing to more complex types of thinking, such as analysis and evaluation. Bloom's Taxonomy levels help faculty to guide students through the learning process, from fundamental remembering and understanding to more complex evaluating and creating.



At KIOT, Curriculum Design, Delivery and Assessment (CDDA) are carried out based on the Blooms' Taxonomy Levels (BTL). Its organized set of objectives helps teachers to plan and deliver appropriate instruction, design valid assessment tasks and schemes, and ensure that instruction and assessment are aligned with the objectives.

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(B) BLOOM'S TAXONOMY ACTION VERBS

I. Remembering	II. Understanding	III. Applying	IV. Analyzing	V. Evaluating	VI. Creating
Exhibit memory of previously learned material by recalling facts, terms, basic concepts, and answers.	Demonstrate understanding of facts and ideas by organizing, comparing, interpreting, giving descriptions, and stating main ideas.	Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations.	Present and defend opinions by making judgments about information, validity of ideas, or quality of work based on a set of criteria.	Compile information together in a different way by combining elements in a new pattern or proposing new Solutions.
- Choose - Define - Find - How - Label - List - Match - Name - Omit - Recall - Relate - Select - Show - Spell - Tell - What - When - Where - Which - Who - Why	- Classify - Compare - Contrast - Demonstrate - Explain - Extend - Illustrate - Infer - Interpret - Outline - Relate - Rephrase - Show - Summarize - Translate	- Apply - Build - Choose - Construct - Develop - Experiment with - Identify - Interview - Make use of - Model - Organize - Plan - Select - Solve - Utilize	- Analyze - Assume - Categorize - Classify - Compare - Conclusion - Contrast - Discover - Dissect - Distinguish - Divide - Examine - Function - Inference - Inspect - List - Motive - Relationships - Simplify - Survey - Take part in - Test for - Theme	- Agree - Appraise - Assess - Award - Choose - Compare - Conclude - Criteria - Criticize - Decide - Deduct - Defend - Determine - Disprove - Estimate - Evaluate - Explain - Importance - Influence - Interpret - Judge - Justify - Mark - Measure - Opinion - Perceive - Prioritize - Prove - Rate - Recommend - Rule on - Select - Support - Value	- Adapt - Build - Change - Choose - Combine - Compile - Compose - Construct - Create - Delete - Design - Develop - Discuss - Elaborate - Estimate - Formulate - Happen - Imagine - Improve - Invent - Make up - Maximize - Minimize - Modify - Original - Originate - Plan - Predict - Propose - Solution - Solve - Suppose - Test - Theory

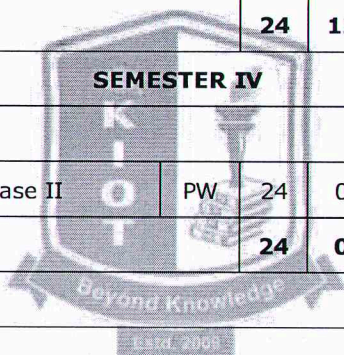
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KNOWLEDGE INSTITUTE OF TECHNOLOGY (AUTONOMOUS), SALEM - 637504											
M.E. INDUSTRIAL SAFETY ENGINEERING									Version : 2.1		
Courses of Study and Scheme of Assessment (Regulations 2023-R1)									Date: 15.12.2025		
S.No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
SEMESTER I											
	THEORY										
1	ME23MA101	Probability and Statistical Methods	FC	4	3	1	0	4	40	60	100
2	UX23RM201	Research Methodology and IPR	RM	3	2	1	0	3	40	60	100
3	ME23IS301	Principles of Safety Management	PC	3	3	0	0	3	40	60	100
4	ME23IS302	Environmental Safety	PC	3	3	0	0	3	40	60	100
5	ME23IS303	Occupational Health and Industrial Hygiene	PC	3	3	0	0	3	40	60	100
6	ME23IS304	Industrial Safety, Health and Environment Acts	PC	3	3	0	0	3	40	60	100
7	ME23IS305	Fire Engineering and Explosion Control	PC	3	3	0	0	3	40	60	100
	PRACTICAL										
8	ME23IS306	Industrial Safety and Simulation Laboratory	PC	2	0	0	2	1	60	40	100
	EMPLOYABILITY ENHANCEMENT										
9	UX23PT801	Technical Seminar / Case Study Presentation	EEC	2	0	0	2	NC	100	-	100
Total				26	20	2	4	23	440	460	900
SEMESTER II											
	THEORY										
1	ME23IS307	System Simulation and Hazard Analysis	PC	4	4	0	0	4	40	60	100
2	ME23IS308	Safety in Process Industries	PC	3	3	0	0	3	40	60	100
3	ME23IS4XX	Professional Elective-I	PE	3	3	0	0	3	40	60	100
4	ME23IS4XX	Professional Elective-II	PE	3	3	0	0	3	40	60	100
5	ME23XX5XX	Open Elective-I	OE	3	3	0	0	3	40	60	100
6	UX23MC701	Universal Human Values and Ethics	MC	3	1	0	2	NC	100	-	100
7	UX23AC7XX	Audit Course	AC	2	2	0	0	NC	100	-	100
	EMPLOYABILITY ENHANCEMENT										
8.	UX23PT802	Research Paper Review and Presentation	EEC	2	0	0	2	1	100	-	100
9.	ME23PT803	Industrial Safety Assessment – Internship	EEC	4	0	0	4	2	100	-	100
Total				27	19	0	8	19	600	300	900

w.e.f. 22/12/2025
 Passed in BoS Meeting held on 08/12/2025
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			CAT	CP	L	T	P	C	IA	ESE	Total	
SEMESTER III												
	THEORY											
1.	ME23IS309	Electrical Safety	PC	3	3	0	0	3	40	60	100	
2.	ME23IS4XX	Professional Elective-III	PE	3	3	0	0	3	40	60	100	
3.	ME23IS4XX	Professional Elective- IV	PE	3	3	0	0	3	40	60	100	
4.	ME23XX5XX	Open Elective-II	OE	3	3	0	0	3	40	60	100	
	PRACTICAL											
5.	ME23IS601	Project Work – Phase I	PW	12	0	0	12	6	60	40	100	
Total				24	12	0	12	18	220	280	500	
SEMESTER IV												
	PRACTICAL											
1.	ME23IS602	Project Work – Phase II	PW	24	0	0	24	12	60	40	100	
Total				24	0	0	24	12	60	40	100	
												Total Number of Credits:72




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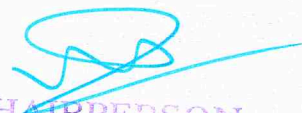
SEMESTER WISE CREDITS DISTRIBUTION

SUMMARY							
S.No.	Course Category	Credits per Semester				Credits	Credit %
		I	II	III	IV		
1.	FC	4	-	-	-	4	5.55
2.	RM	3	-	-	-	3	4.16
3.	PC	16	7	3	-	26	36.11
4.	PE	-	6	6	-	12	16.66
5.	OE	-	3	3	-	6	8.33
6.	PW	-	-	6	12	18	25.0
7.	AC/MC	✓	✓	-	-	0	0
8.	EEC	-	3	-	-	3	4.16
	Total	24	19	18	12	72	100

NOMENCLATURE

CAT	Category of Course	FC	Foundation Courses	AC/MC	Audit Courses / Mandatory Courses
CP	Contact Periods	RM	Research Methodology and IPR	EEC	Employability Enhancement Courses
L	Lecture Periods	PC	Professional Core Courses	IA	Internal Assessment
T	Tutorial Periods	PE	Professional Elective Courses	ESE	End Semester Examination
P	Laboratory Periods	OE	Open Elective Courses		
C	Credits	PW	Project Work Courses		

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PROFESSIONAL ELECTIVES											
SEMESTER II (Professional Electives - I & II)											
S. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
1.	ME23IS401	Plant Layout and Material Handling	PE	3	3	0	0	3	40	60	100
2.	ME23IS402	Work Study and Ergonomics	PE	3	3	0	0	3	40	60	100
3.	ME23IS403	Human Factors in Engineering	PE	3	3	0	0	3	40	60	100
4.	ME23IS404	Maintenance Engineering	PE	3	3	0	0	3	40	60	100
5.	ME23IS405	Optimization Techniques	PE	3	3	0	0	3	40	60	100
6.	ME23IS406	Transport Safety	PE	3	3	0	0	3	40	60	100
7.	ME23IS407	Fireworks Safety	PE	3	3	0	0	3	40	60	100
8.	ME23IS408	Nuclear Engineering and Safety	PE	3	3	0	0	3	40	60	100
9.	ME23IS409	Safety in construction	PE	3	3	0	0	3	40	60	100
SEMESTER III (Professional Electives - III & IV)											
1	ME23IS410	Safety in Textile Industry	PE	3	3	0	0	3	40	60	100
2	ME23IS411	Safety in Mines	PE	3	3	0	0	3	40	60	100
3	ME23IS412	Dock Safety	PE	3	3	0	0	3	40	60	100
4	ME23IS413	Safety in Engineering Industry	PE	3	3	0	0	3	40	60	100
5	ME23IS414	Quality Engineering in Production Systems	PE	3	3	0	0	3	40	60	100
6	ME23IS415	ISO 45001 and ISO 14000	PE	3	3	0	0	3	40	60	100
7	ME23IS416	Artificial Intelligence and Data Analytics	PE	3	3	0	0	3	40	60	100
8	ME23IS417	Design of Experiments	PE	3	3	0	0	3	40	60	100
9	ME23IS418	Reliability Engineering	PE	3	3	0	0	3	40	60	100
10	ME23IS419	Logistics and Distribution Management	PE	3	3	0	0	3	40	60	100

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OPEN ELECTIVES													
Sl. No.	Course Code	Course Title	Offered by	Periods / Week						Maximum Marks			
				CAT	CP	L	T	P	C	IA	ESE	Total	
Except M.E. Industrial Safety Engineering													
1	ME23IS501	Environmental Safety	MECH	OE	3	3	0	0	3	40	60	100	
2	ME23IS502	Electrical Safety	MECH	OE	3	3	0	0	3	40	60	100	
3	ME23IS503	Safety in Engineering Industry	MECH	OE	3	3	0	0	3	40	60	100	
4	ME23IS504	Design of Experiments	MECH	OE	3	3	0	0	3	40	60	100	
5	ME23IS505	Circular Economy	MECH	OE	3	3	0	0	3	40	60	100	
Except M.E. Automotive Electronics													
6	ME23AE501	Fundamentals of Automotive Electronics	ECE	OE	3	3	0	0	3	40	60	100	
7	ME23AE502	Vehicle Sensors and Electronic Control Systems	ECE	OE	3	3	0	0	3	40	60	100	
8	ME23AE503	Automotive Embedded Systems	ECE	OE	3	3	0	0	3	40	60	100	
9	ME23AE504	Automotive Communication Networks	ECE	OE	3	3	0	0	3	40	60	100	
10	ME23AE505	Introduction to Electric and Hybrid Vehicle Electronics	ECE	OE	3	3	0	0	3	40	60	100	
Except M.E. Power Electronics and Drives													
11	ME23PE501	Electric Vehicle Technology	EEE	OE	3	3	0	0	3	40	60	100	
12	ME23PE502	Energy Conservation and Management	EEE	OE	3	3	0	0	3	40	60	100	
13	ME23PE503	Smart Grid	EEE	OE	3	3	0	0	3	40	60	100	
14	ME23PE504	Solar PV and Energy Storage Systems	EEE	OE	3	3	0	0	3	40	60	100	
Except M.E. Software Engineering													
15	ME23SE501	Blockchain Technologies	CSE	OE	3	3	0	0	3	40	60	100	
16	ME23SE502	Deep Learning	CSE	OE	3	3	0	0	3	40	60	100	
17	ME23SE503	Full Stack Web Application Development	CSE	OE	3	3	0	0	3	40	60	100	
18	ME23SE504	Principles of Multimedia	CSE	OE	3	3	0	0	3	40	60	100	
19	ME23SE505	Cyber Physical Systems	CSE	OE	3	3	0	0	3	40	60	100	
AUDIT COURSES													
1	UX23AC701	Industrial Safety Engineering	MECH	AC	2	2	0	0	0	100	-	100	
2	UX23AC702	Sustainability Engineering	MECH	AC	2	2	0	0	0	100	-	100	
3	UX23AC703	Energy Audit	EEE	AC	2	2	0	0	0	100	-	100	
4	UX23AC704	Design Thinking	CSE	AC	2	2	0	0	0	100	-	100	
5	UX23AC705	Business Analytics	CSE	AC	2	2	0	0	0	100	-	100	

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M.E./ M.Tech. / M.C.A. Regulations 2023

ME23MA101	PROBABILITY AND STATISTICAL METHODS	CP	L	T	P	C
		4	3	1	0	4
Programme & Branch	M.E. INDUSTRIAL SAFETY ENGINEERING	Version: 1.0				
Use of statistical tables are permitted						
Course Objectives:						
1.	To introduce the basic concepts of probability and standard distributions.					
2.	To provide the most appropriate estimator of the parameter in statistical inference.					
3.	To decide whether to accept or reject a specific value of a parameters.					
4.	To introduce the fundamentals of classifications of design of experiments which plays very important roles in the field of agriculture and quality control.					
5.	To learn methods for analyzing time series data to extract meaningful statistical characteristic of data.					
UNIT-I	PROBABILITY AND RANDOM VARIABLES					9+3
	Probability – Axioms of probability – Conditional probability and problems – Baye’s theorem and problems – Random variables – Probability function – Moments – Moment generating functions and their properties – Binomial, Poisson, Geometric, Uniform, Exponential, Gamma and Normal distributions and problems – Function of a random Variable.					
UNIT-II	ESTIMATION THEORY					9+3
	Principle of least squares – Regression and problems – Multiple and partial correlations and problems – Estimation of parameters and problems – Maximum likelihood estimates – Method of moments and problems.					
UNIT- III	TESTING OF HYPOTHESIS					9+3
	Sampling distributions – Small and large samples and problems – Tests based on Normal, t-distribution, Chi - square, Goodness of fit and F - distributions.					
UNIT IV	DESIGN OF EXPERIMENTS					9+3
	Analysis of variance – Completely randomized design – Randomized block design – Latin square design – 2 ² Factorial designs.					
UNIT-V	TIME SERIES					9+3
	Characteristics and representation - Moving averages and problems – Exponential smoothing – Auto Regressive Processes and problems.					
	Total(LT)					60 Periods

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M.E./ M.Tech. / M.C.A. Regulations 2023
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OPEN ENDED PROBLEMS / QUESTIONS

Course specific open-ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.

COURSE OUTCOMES:

Upon completion of this course the students will be able to:

**BLOOM'S
Taxonomy**

CO1	Explain probability axioms, rules and the moments of discrete and continuous random variables.	L2 - Understand
CO2	Interpret the concepts of estimator and various methods to find Estimator.	L2 - Understand
CO3	Utilize statistical tests in testing hypotheses on data.	L3 - Apply
CO4	Apply the basic concepts of classifications of design of experiments in the field of agriculture and quality control.	L3 - Apply
CO5	Explain various time series models and application of these models appropriately to engineering problems.	L2 - Understand

REFERENCE BOOKS:

1.	Anderson, O.D, "Time Series Analysis: Theory and Practice", North - Holland, Amsterdam, 1982.
2.	Devore, J. L., "Probability and Statistics for Engineering and Sciences", 9th Edition, Cengage Learning, 2016.
3.	Gupta S.C. and Kapoor V.K., "Fundamentals of Mathematical Statistics", 12th Edition, Sultan and Sons, New Delhi, 2020.
4.	Johnson, R.A., Miller, I and Freund J., "Miller and Freund" Probability and Statistics for Engineers, 9th Edition, Pearson Education, Asia, 2016.

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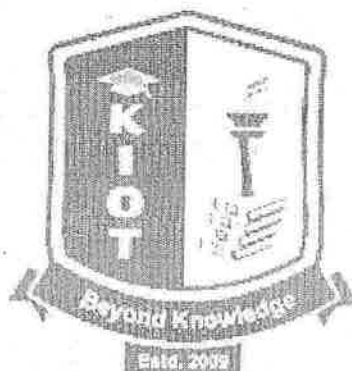
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Mapping of COs with POs						
CO	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	2	2		1
CO2	2	1	2	1		1
CO3	3	2	3	2	1	2
CO4	3	2	3	3	1	2
CO5	3	2	1	3		2
Average	2.6	1.6	2.2	2.2	0.4	1.6
1-Low, 2 -Medium, 3-High.						


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UX23RM201	RESEARCH METHODOLOGY AND IPR	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	Common to all M.E (SE,AE,PED,ISE) and M.C.A	Version: 1.0				
Course Objectives:						
1.	To analyze the significance of research and formulate well-defined research questions.					
2.	To apply appropriate research methods and critically evaluate research articles.					
3.	To create well-structured research papers and utilize research tools proficiently.					
4.	To produce effective technical reports and deliver impactful presentations.					
5.	To understand forms of intellectual property and analyze their implications on technological research and international cooperation.					
UNIT-I	CONCEPT OF RESEARCH					6+3
	Meaning and Significance of Research-Skills, Habits and Attitudes for Research-Time Management -Status of Research in India - Why, How, and What a Research is?-Types and Process of Research-Outcome of Research-Sources of Research Problem-Characteristics of a Good Research Problem -Errors in Selecting a Research Problem-Importance of Keywords -Literature Collection - Analysis -Citation Study - Gap Analysis-Problem Formulation Techniques.					
UNIT-II	RESEARCH METHODS AND JOURNALS					6+3
	Interdisciplinary Research-Need for Experimental Investigations - Data Collection Methods -Appropriate Choice of Algorithms / Methodologies / Methods -Measurement and Result Analysis-Investigation of Solutions for Research Problem-Interpretation - Research Limitations-Journals in Science/Engineering-Indexing and Impact factor of Journals -Citations- h Index- i10 Index -Journal Policies, How to Read a Published Paper-Ethical Issues Related to Publishing- Plagiarism and Self-Plagiarism.					
UNIT-III	PAPER WRITING AND RESEARCH TOOLS					6+3
	Types of Research Papers - Original Article/Review Paper/Short Communication/Case Study-When and Where to Publish? - Journal Selection Methods-Layout of a Research Paper-Guidelines for Submitting the Research Paper-Review Process - Addressing Reviewer Comments-Use of tools / Techniques for Research -Hands-on Training related to Reference Management Software - EndNote - Introduction to Origin, SPSS,etc-Software for Detection of Plagiarism.					
UNIT - IV	EFFECTIVE TECHNICAL THESIS WRITING/PRESENTATION					6+3
	How to Write a Report- - Language and Style-Format of Project Report - Use of Quotations-Method of Transcription Special Elements -Title Page - Abstract - Table of Contents - Headings and Sub-Headings -Footnotes - Tables and Figures - Appendix - Bibliography					

w.e.f. 13/10/2025

Passed in BoS Meeting held on 05/09/2023

Approved in Academic Council Meeting held on 11/10/2025

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	etc. -Different Reference Formats -Presentation using PPTs.	
UNIT – V	NATURE OF INTELLECTUAL PROPERTY	6+3
	Patents - Designs - Trade and Copyright - Process of Patenting and Development - Technological research- innovation- patenting- Development International Scenario -International Cooperation on Intellectual Property -Procedure for Grants of Patents.	
	Total(LT)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Illustrate the importance and objectives of research in contributing to knowledge and solving real-world problems.	L2 - Understand
CO2	Experiment with data collection techniques, choosing fitting approaches to ensure sound research framework and methodology.	L3 - Apply
CO3	Utilize research & analytic tools for enhancing the research publication	L2 - Understand
CO4	Apply knowledge to produce presentations and technical reports that effectively communicate research findings.	L3 - Apply
CO5	Explain types of intellectual property and comprehend patenting as essential for safeguarding innovation and creativity.	L2 - Understand
	REFERENCE BOOKS:	
1.	Cooper, Donald R.; Schindler, Pamela S.; Sharma, J. K., Business Research Methods, 12 th Edition, Tata McGraw Hill Education, 2012.	
2.	DePoy, Elizabeth; Gitlin, Laura N., Introduction to Research: Understanding and Applying Multiple Strategies, 6 th Edition, Elsevier Health Sciences, 2015.	
3.	Walliman, Nicholas, Research Methods: The Basics, 3 rd Edition, Routledge, 2017.	
4.	Bettig, Ronald V., Copyrighting Culture: The Political Economy of Intellectual Property, 1 st Edition, Routledge, 2018.	
5.	The Institute of Company Secretaries of India, Statutory body under an Act of parliament, "Professional Programme Intellectual Property Rights, Law and practice", September 2013.	

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	2	1			1
CO2	3	3		2		
CO3	3			3	1	
CO4	3	3				
CO5	2	2		2		1
Average	2.8	2.5	1	2.33	1	1
1-Low, 2 -Medium, 3-High.						




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ME23IS301	PRINCIPLES OF SAFETY MANAGEMENT	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. INDUSTRIAL SAFETY ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To achieve an understanding of principles of safety management.					
2.	To enable the students to learn about various functions and activities of safety department.					
3.	To enable students to conduct safety audit and write audit reports effectively in auditing situations.					
4.	To have knowledge about sources of information for safety promotion and training.					
5.	To familiarize students with evaluation of safety performance.					
UNIT-I	CONCEPTS AND TECHNIQUES					9
	History of Safety Movement - Evolution of Modern Safety Concept - General Concepts of Management - Planning for Safety for Optimization of Productivity - Productivity, Quality, and Safety - Line and Staff Functions for Safety - Budgeting for Safety - Safety Policy - Incident Recall Technique (IRT) - Disaster Control - Job Safety Analysis -Safety Survey -Safety Inspection - Safety Sampling - Evaluation of Performance of Supervisors on Safety.					
UNIT-II	SAFETY EDUCATION AND TRAINING					9
	Importance of Training - Assessing competency - Identification of Training Needs - Training Methods - Programmed, Seminars - Conferences - Competitions - Method of Promoting Safe Practice - Motivation - Communication - Role of Government Agencies and Private Consulting Agencies in Safety Training - Creating Awareness - Awards, Celebrations, Safety Posters, Safety Displays, Safety Pledge, Safety Incentive Scheme, Safety Campaign - Domestic Safety and Training.					
UNIT-III	SAFETY AUDIT					9
	Components of Safety Audit - Types of Audits - Audit Methodology - Non-Conformity Reporting (NCR) - Audit Checklist and Report - Review of Inspection - Remarks by Government Agencies, Consultants, Experts -Perusal of Accident and Safety Records, Formats - Implementation of Audit Indication - Liaison with Departments to Ensure Coordination - Checklist -Identification of Unsafe Acts of Workers and Unsafe Conditions in the Shop Floor.					
UNIT-IV	ACCIDENT INVESTIGATION AND REPORTING					9
	Concept of an Accident - Reportable and Non-reportable Accidents -Reporting to Statutory Authorities - Principles of Accident Prevention -Accident Investigation and Analysis - Records for Accidents, Departmental Accident Reports, Documentation of Accidents - Unsafe Act and Condition -Domino Sequence - Supervisory Role - Role of Safety Committee - Cost of Accident.					
UNIT - V	SAFETY PERFORMANCE MONITORING					9
	ANSI (Z16.1) Recommended Practices for Compiling and Measuring Work Injury Experience - Permanent Total Disabilities - Permanent Partial Disabilities - Temporary Total Disabilities - Calculation of Accident Indices - Frequency Rate, Severity Rate, Frequency Severity Incidence, Incident Rate,					

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	Accident Rate, Safety "t" Score, Safety Activity Rate -Problems - Financial justification of hazard controls.	
	Total (L)	45 Periods

OPEN ENDED PROBLEMS / QUESTIONS

Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.

COURSE OUTCOMES:

Upon completion of this course the students will be able to:

BLOOM'S Taxonomy

CO1	Summarize the various concepts and Techniques in the safety management.	L2 - Understand
CO2	Explain the safety education and training.	L2 - Understand
CO3	Organize a safety audit and prepare a report for the audit.	L3 - Apply
CO4	Develop an accident investigation report.	L3 - Apply
CO5	Examine the safety performance monitoring activities.	L4 - Analyze

REFERENCE BOOKS:

1.	"Accident Prevention Manual for Industrial Operations", N.S.C.Chicago, 13th Edition 2009.
2.	Blake R.B., "Industrial Safety" Prentice Hall, Inc., New Jersey,. 3 rd Edition 2000.
3.	Dan Petersen, "Techniques of Safety Management", McGraw-Hill Company, Tokyo, 1981.
4.	Heinrich H.W. "Industrial Accident Prevention" McGraw-Hill Company, New York, 1980
5.	John Ridley, "Safety at Work", Butterworth and Co., London, 1983
6.	Lees, F.P., "Loss Prevention in Process Industries" Butterworth publications, London, 2 nd edition, 1990.
7.	Relevant Indian Standards and Specifications, BIS, New Delhi. 8. "Safety and Good House Keeping", N.P.C., New Delhi, 1985.

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1		1		2	2	
CO2				2	3	
CO3	3	3	2	2	1	1
CO4	2	3	2	3	3	
CO5			3	3		2
Average	2.5	2.3	2.3	2.4	2.25	1.2
1-Low, 2 -Medium, 3-High.						



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ME23IS302		ENVIRONMENTAL SAFETY		CP	L	T	P	C
Programme & Branch		M.E. INDUSTRIAL SAFETY ENGINEERING		3	3	0	0	3
				Version: 1.0				
Course Objectives:								
1.	To provide in depth knowledge in Principles of Environmental safety and its applications in various fields.							
2.	To give understanding of air and water pollution and their control.							
3.	To expose the students to the basis in hazardous waste management.							
4.	To provide knowledge on pollution monitoring and control devices.							
5.	To design emission measurement devices.							
UNIT-I	AIR POLLUTION							9
	Classification and properties of air pollutants - Pollution sources - Effects of air pollutants on human beings - Animals, Plants, and Materials - Automobile pollution - Hazards of air pollution - Concept of clean coal combustion technology - Ultra violet radiation - infrared radiation - radiation from the sun - Hazards due to depletion of ozone - Deforestation - ozone holes - Automobile exhausts, chemical factory stack emissions, CFC.							
UNIT-II	WATER POLLUTION							9
	Classification of water pollutants - Health hazards - Sampling and analysis of water - Water treatment - Different industrial effluents and their treatment and disposal - Advanced wastewater treatment - Effluent quality standards and laws - Chemical industries, tannery, textile effluents - Common treatment.							
UNIT- III	HAZARDOUS WASTE MANAGEMENT							9
	Hazardous waste management in India - Waste identification, characterization, and classification - Technological options for collection, treatment, and disposal of hazardous waste - Selection charts for the treatment of different hazardous wastes - Methods of collection and disposal of solid wastes - Health hazards - Toxic and radioactive wastes - Incineration and verification - Hazards due to bio-process - dilution, standards, and restrictions - Recycling and reuse.							
UNIT - IV	ENVIRONMENTAL MEASUREMENT AND CONTROL							9
	Sampling and analysis - Dust monitor - Gas analyzer - particle size analyzer - Lux meter - pH meter - Gas chromatograph - atomic absorption spectrometer - Gravitational settling chambers - cyclone separators - scrubbers - Electrostatic precipitator, bag filter, maintenance - Control of gaseous emission by adsorption, absorption, and combustion methods - Pollution Control Board, laws.							
UNIT-V	POLLUTION CONTROL IN PROCESS INDUSTRIES							9
	Pollution control in process industries - Cement, paper, petroleum, petroleum products, textile - Tanneries, thermal power plants - Dyeing and pigment industries - Eco-friendly energy.							

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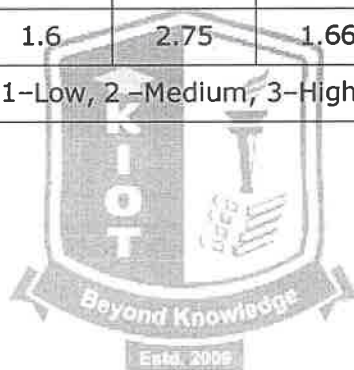
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		Total (L)	45 Periods
OPEN ENDED PROBLEMS / QUESTIONS			
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.			
COURSE OUTCOMES: Upon completion of this course the students will be able to:			BLOOM'S Taxonomy
CO1	Illustrate and familiarize the basic concepts scope of environmental safety.	L2 - Understand	
CO2	Interpret the standards of professional conduct that are published by professional safety organizations and/or certification bodies.	L2 - Understand	
CO3	Explain the ways in which environmental health problems have arisen due to air and water pollution.	L2 - Understand	
CO4	Examine the role of hazardous waste management and use of critical thinking to identify and assess environmental health risks.	L4 - Analyze	
CO5	Apply concepts of emission measurement and design emission measurement devices.	L3 - Apply	
REFERENCE BOOKS:			
1.	E. C Wolfe, Race to Save to Save Planet, Wadsworth Publishing Co., Belmont, CA 2006.		
2.	G. T Miller, Environmental Science: Working with the Earth, 11th Edition, Wadsworth Publishing Co., Belmont, CA, 2006		
3.	M.J Hammer,, and M.J Hammer,, Jr., Water and Wastewater Technology, Pearson Prentice Hall, 2006		
4.	Rao, CS, "Environmental pollution engineering: Wiley Eastern Limited, New Delhi, 1 st January 2018.		
5.	S. P. Mahajan, "Pollution control in process industries", Tata McGraw Hill Publishing Company, New Delhi, 2006.		
6.	Varma and Braner, "Air pollution equipment", Springer Publishers, Second Edition.		

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1		1	2			1
CO2		1	3	1		1
CO3		2	2		1	2
CO4	1	3	3	1	3	
CO5	1	1	3	3		
Average	1	1.6	2.75	1.66	2	1.33
1-Low, 2 -Medium, 3-High.						



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ME23IS303	OCCUPATIONAL HEALTH AND INDUSTRIAL HYGIENE	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E INDUSTRIAL SAFETY ENGINEERING	Version: 1.0				
Course Objectives:						
1	To apply the knowledge of physical hazards and its control measures in an Industrial Environment.					
2	To distinguish the types of chemicals for its health hazard and provide suitable control methods					
3	To categorize various types of hazards arising out of biological and ergonomical aspects in a process and able to provide suitable corrective actions					
4	To implement the functions and activities of Occupational health services.					
5	To illustrate the various physiological functions of our body and the test methods for periodical monitoring of health.					
UNIT-I	PHYSICAL HAZARDS					9
	General physics concepts (e.g., force, acceleration, velocity, momentum, and friction) – Noise : compensation aspects, noise exposure regulation, properties of sound, occupational damage, risk factors, sound measuring instruments , octave band analyzer, noise networks, noise surveys, noise control program, industrial audiometry, hearing conservation programs – Vibration : types, effects, instruments, surveying procedure, permissible exposure limit - Ionizing radiation : types, effects, monitoring instruments, control programs, OSHA standard - Non-ionizing radiations : effects, types, radiation hazards, microwaves and radio-waves, lasers, TLV - Cold environments, hypothermia, wind chill index, control measures - Hot environments, thermal comfort, heat stress indices, acclimatization, estimation, and control.					
UNIT-II	CHEMICAL HAZARDS					9
	General chemistry concepts (e.g., nomenclature, balancing chemical equations, chemical reactions, ideal gas law, and pH) - Recognition of chemical hazards: dust, fumes, mist, Vapor, fog, gases, types, concentration Exposure vs. dose, TLV - Methods of Evaluation: process or operation description, Field Survey, Sampling methodology, Industrial Hygiene calculations, Comparison with OSHAS Standard, Air Sampling instruments , Types, Measurement Procedures, Instruments Procedures, Gas and Vapour monitors, dust sample collection devices, personal sampling, Methods of Control: Engineering Control, Design maintenance considerations, design specifications, General Control Methods, training, and education.					
UNIT- III	BIOLOGICAL AND ERGONOMICAL HAZARDS					9
	Classification of Bio hazardous agents: examples, bacterial agents - rickettsial and chlamydial agents - viral agents, fungal, parasitic agents, infectious diseases - Protocol for blood borne pathogen control – Covid, SARS - Protocol for Air borne pathogen control - Biohazard control program , employee health program , laboratory safety program, animal care and handling, biological safety cabinets , building design- Work Related Musculoskeletal Disorders : carpal tunnel syndrome CTS, Tendon pain, disorders of the neck, back injuries.					

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UNIT – IV	OCCUPATIONAL HEALTH AND TOXICOLOGY	9
	Concept and spectrum of health, Functional units and activities of occupational health services, pre-employment, and post-employment medical examinations, Occupational related diseases, levels of prevention of diseases, Notifiable occupational diseases: such as silicosis, asbestosis, pneumoconiosis, siderosis, anthracosis, aluminosis, and anthrax, Lead-nickel, chromium, and manganese toxicity, gas poisoning (such as CO, ammonia, coal, and dust, etc.), their effects and prevention, Cardio pulmonary resuscitation, audiometric tests, eye tests, vital function tests, Industrial toxicology: local, systemic, and chronic effects, temporary and cumulative effects, Mutagens, teratogens, and carcinogens, carcinogens entry into human systems.	
UNIT–V	OCCUPATIONAL PHYSIOLOGY	9
	Man as a system component - allocation of functions - efficiency- occupational work capacity - Aerobic and anaerobic work - evaluation of physiological requirements of jobs - parameters of measurements - Categorization of job heaviness - work organization - stress - strain - fatigue - rest pauses - shift work - personal hygiene.	
Total (L)		45 Periods

OPEN ENDED PROBLEMS / QUESTIONS

Course specific open-ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination

COURSE OUTCOMES:

Upon completion of this course the students will be able to:

BLOOM'S Taxonomy

CO1	Apply the knowledge of physical hazards and its control measures in an Industrial Environment	L3 - Apply
CO2	Explain the types of chemicals for its health hazard and provide suitable control methods	L2 Understand
CO3	Interpret various types of hazards arising out of biological and ergonomical aspects in a process and able to provide suitable corrective actions	L2 Understand
CO4	Outline the functions and activities of Occupational health services.	L2 Understand
CO5	Evaluate the various physiological functions of our body and the test methods for periodical monitoring of health.	L5 - Evaluate

REFERENCE BOOKS:

1. Benjamin O.Alli, Fundamental Principles of Occupational Health and Safety ILO 2008.
2. Danuta Koradecka, Handbook of Occupational Health and Safety, CRC, 2010.
3. E.J. McCornick, and M. S Sanders, Human Factors in Engineering and Design, Tata McGraw-Hill, 1992.

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4.	Encyclopedia of "Occupational Health and Safety", Vol.I and II, published by International Labour Office, Geneva, 1985
5.	Hand book of "Occupational Safety and Health", National Safety Council, Chicago, 2002.
6.	Lawrence Slote, Handbook of occupational safety and health, Wiley, 2001.
7.	Louis J. di Berardinis, Handbook of occupational safety and health Wiley, 1999.
8.	Interim guidance "COVID-19: Occupational health and safety for health workers", WHO & ILO, 2021

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1	1	3	2	2	
CO2	1	1	3	2	2	
CO3		1	3		2	1
CO4		1	3		1	1
CO5			2		1	1
Average	1	1	2.8	2	1.6	1
1-Low, 2-Medium, 3-High.						

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UNIT –IV	OTHER ACTS AND RULES	9
	Indian Boiler (Amendments) Act 2007 - static and mobile pressure vessel rules (SMPV) - motor vehicle rules, The Mines and Minerals (Development & Regulation) Amendment Act 2015 - workman compensation act, rules - electricity act and rules - hazardous wastes (management, handling and transboundary) rules, 2008 - the building and other construction workers act 1996, Petroleum rules, Gas cylinder rules 2016 - Explosives Act 1884 - Pesticides Act - E waste (management) rules 2016.	
UNIT–V	INTERNATIONAL ACTS AND STANDARDS	9
	Occupational Safety and Health act of USA (The Williams - Steiger Act of 1970) - Health and safety work act (HASAWA 1974, UK) - ISO 14001 - ISO 45001, European Safety and Health Legislations, American Petroleum Institute (API) Standards, Oil Industry Safety Directorate (OISD) Standards, National Fire Protection Association (NFPA) Standards, Atomic Energy Regulatory Board (AERB), American National Standards Institute (ANSI).	
	Total (L)	45 Periods

OPEN ENDED PROBLEMS / QUESTIONS

Course specific open-ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.

COURSE OUTCOMES: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
C01	Interpret the requirements mentioned in factories act for the prevention of accidents.	L2 - Understand
C02	List important legislations related to health, Safety and Environment act.	L1- Remember
C03	Infer the manufacturing storage and import of hazardous chemical rule.	L2 - Understand
C04	Summarize the statutory requirements for an Industry on registration, license and its renewal.	L2 - Understand
C05	Develop a system of international act and standard.	L3- Apply
REFERENCE BOOKS:		
1.	The Factories Act 1948, Madras Book Agency, Chennai, 2000	
2.	The Environment Act (Protection) 1986, Commercial Law Publishers (India) Pvt.Ltd., New Delhi.	
3.	Water (Prevention and control of pollution) act 1974, Commercial Law publishers (India) Pvt.Ltd., New Delhi.	
4.	Air (Prevention and control of pollution) act 1981, Commercial Law Publishers (India) Pvt.Ltd., New Delhi.	
5.	The Indian boilers act 1923, Commercial Law Publishers (India) Pvt.Ltd., Allahabad.	

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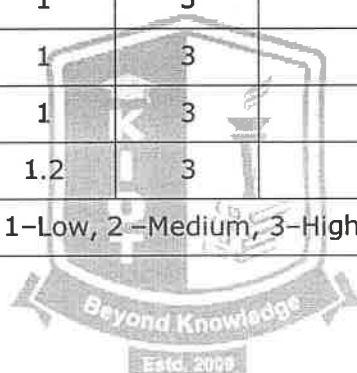
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6.	The Mines Act 1952, Commercial Law Publishers (India) Pvt.Ltd., Allahabad.
7.	The manufacture, storage and import of hazardous chemical rules 1989, Madras Book Agency, Chennai.
8.	Srinivasan S , "The Tamil Nadu Safety Officers Rules 2005" Madras Book Agency, Chennai, 28th Edition, 2017

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1		2	3		2	
CO2	1	1	3		2	2
CO3	1	1	3		2	1
CO4		1	3		2	1
CO5		1	3		2	
Average	1	1.2	3		2	1.33
1-Low, 2-Medium, 3-High.						



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ME23IS305	FIRE ENGINEERING AND EXPLOSION CONTROL	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E INDUSTRIAL SAFETY ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To provide an in-depth knowledge about the science of fire.					
2.	To understand the causes and effects of fire.					
3.	To know the various fire prevention systems and protective equipment's.					
4.	To understand the science of explosion and its prevention techniques.					
5.	To understand the various fire prevention techniques to be followed in a building.					
UNIT-I	PHYSICS AND CHEMISTRY OF FIRE	9				
	Fire properties of solid, liquid and gases - fire spread- toxicity of products of combustion - theory of combustion and explosion- vapour clouds - Combustible dust - flash fire - jet fires - pool fires - unconfined vapour cloud explosion, shock waves - auto-ignition - boiling liquid expanding vapour explosion - case studies - Flix borough, Mexico disaster, Pasedena Texas, Piper Alpha, Peterborough and Bombay Victoria dock ship explosions.					
UNIT-II	FIRE PREVENTION AND PROTECTION	9				
	Sources of ignition - fire triangle - principles of fire extinguishing - active and passive fire protection systems - various classes of fires - A, B, C, D, E - types of fire extinguishers - fire stoppers - hydrant pipes - hoses - monitors - fire watchers - lay out of stand pipes - fire station - fire alarms and sirens - maintenance of fire trucks - foam generators - escape from fire rescue operations - fire drills - notice-first aid for burns.					
UNIT- III	INDUSTRIAL FIRE PROTECTION SYSTEMS	9				
	Sprinkler -hydrants - stand pipes - special fire suppression systems like deluge and emulsifier, selection criteria of the above installations, reliability, maintenance, evaluation and standards - alarm and detection systems. Other suppression systems - CO2 system, foam system, dry chemical powder (DCP) system, Halon system - need for halon replacement - smoke venting. Portable extinguishers - flammable liquids - tank farms - indices of in flammability - firefighting systems.					
UNIT - IV	BUILDING FIRE SAFETY	9				
	Objectives of fire safe building design, Fire load, fire resistant material and fire testing - structural fire protection - structural integrity - concept of egress design - exists - width calculations - fire certificates - fire safety requirements for high rise buildings - snookers.					
UNIT-V	EXPLOSION PROTECTING SYSTEMS	9				

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	Principles of explosion - detonation and blast waves - explosion parameters - Explosion Protection, Containment, Flame Arrestors, isolation, suppression, venting, explosion relief of large enclosure - explosion venting- inert gases, plant for generation of inert gas - rupture disc in process vessels and lines explosion, suppression system based on carbon dioxide (CO ₂) and halons - hazards in LPG, ammonia (NH ₃), Sulphur dioxide (SO ₃), chlorine (CL ₂) etc.	
	Total (L)	45 Periods
OPEN ENDED PROBLEMS / QUESTIONS		
Course specific open-ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.		
COURSE OUTCOMES: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Explain the basic concepts of fire and explosion science.	L2 - Understand
CO2	Demonstrate the different source of ignition and their prevention techniques.	L2 - Understand
CO3	Illustrate the operation of various types of firefighting equipment's.	L2 - Understand
CO4	Summarize the causes and prevention of explosion.	L2 - Understand
CO5	Apply explosion protection techniques and their significances to suit the industrial requirement.	L3 - Apply
REFERENCE BOOKS:		
1.	"Accident Prevention manual for industrial operations" N.S.C., Chicago, 1982.	
2.	"Davis Daniel et al, "Hand Book of fire technology"	
3.	"Fire Prevention and firefighting", Loss prevention Association, India.	
4.	Derek, James, "Fire Prevention Hand Book", Butter Worths and Company, London, 1986.	
5.	Dinko Tuhtar, "Fire and explosion protection"	
6.	Fire fighters hazardous materials reference book Fire Prevention in Factories", an Nostrand Rein Hold, New York, 1991.	
7.	Gupta, R.S., "Hand Book of Fire Technology" Orient Longman, Bombay 1977.	
8.	Relevant Indian Acts and rules, Government of India.	

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1		3				2
CO2			3		2	
CO3			2	3		
CO4		3			3	
CO5	2					2
Average	2	3	2.5	3	2.5	2
1-Low, 2 -Medium, 3-High.						



w.e.f. 13/10/2025
 Passed in BoS Meeting held on 05/09/2023
 Approved in Academic Council Meeting held on 11/10/2025


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ME23IS306	INDUSTRIAL SAFETY AND SIMIULATION LABORATORY	CP	L	T	P	C
		2	0	0	2	1
Programme & Branch	M.E. INDUSTRIAL SAFETY ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To provide opportunity to operate the equipment to acquire practical knowledge.					
2.	To know the various PPEs and software.					
3.	To carry out experiments to find out the environmental parameters.					
4.	To assess the impact of sensitivity of chemicals on explosivity.					
5.	To run the software to assess the consequence effects of major accidents. To learn the proper use of various kinds of physics laboratory equipment.					
List of Experiments/Exercises and Skills:						
1.	First aid concepts: Study of Emergency Kits ,First – aid, road safety signs and signals -Safety Software Demo.					
2.	Noise level measurement and analysis: Measurement of sound pressure level in dB for impact, continuous and intermittent sources at various networks, peak and average values.					
3.	Friction test: Explosive materials like barium nitrate, gun powder, white powder, amorces composition etc.					
4.	Impact test: Explosive materials like gun powder, white powder, amerce composition etc. Burst strength test of packaging materials like paper bags, corrugated cartoons, wood etc. Auto ignition temperature test.					
5.	Exhaust gas measurement and analysis: Measurement of sox, nox, cox, hydrocarbons.					
6.	Environmental parameter measurement: Dry bulb temperature, wet bulb temperature, determination of relative humidity, wind flow and effective corrective effective. Particle size measurement, Air sampling analysis.					
7.	Personal protective equipment: Respiratory and non-respiratory-demonstration-self-contained breathing apparatus. Safety helmet,Belt, hand gloves, goggles, safety shoe, gum boots, ankle shoes, face shield, nose mask, ear plug, ear muff, anti-static and conducting plastics/rubber materials, apron and leg guard.					
8.	Fire extinguishers and its operations: Water Co2,Foam,Carbon dioxide (Co2),Dry chemical powder and, Currently amendment fire safety systems.					
9.	Static charge testing: on plastic, rubber, ferrous and non-ferrous materials.					
10.	Illumination testing: - by lux meter and photo meter.					

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11.	Electrical safety: Insulation resistance for motors and cables, Estimation of earth resistance. Earth continuity test, Sensitivity test for MCB, ELCB, RCCB, MCCB.
12.	Software usage: Dispersion modeling of various highly dangerous chemicals using aloha software. Software usage - accident analysis ,safety audit packages, consequence analysis (CISCON), fire, explosion and toxicity index (FETI), reliability analysis for mechanical system and electrical System, failure mode analysis.
13.	Experiments on simulation to be added Discrete and continuous

List of Equipment Required:

1.	Noise level meter	:	1 Number
2.	Friction tester	:	1 Number
3.	Impact tester	:	1 Number
4.	Exhaust gas analyzer	:	1 Number
5.	High volume sampler	:	1 Number
6.	PPE Set	:	1 Number
7.	Fire extinguisher set	:	1 Number
8.	Static charge tester	:	1 Number
9.	First aid kit	:	1 Number
10.	Lock out/Tag out	:	1 Number
11.	Software	:	ALOHA, CAMEO
12.	Extend SIM	:	
13.	System	:	12 Number

COURSE OUTCOMES:

Upon completion of this course the students will be able to:

**BLOOM'S
Taxonomy**

1.	Make use of various equipment's to bring out the safety environment in the industry.	L3 - Apply
2.	Measure the particulate matter and assess the impact of air pollution.	L5 - Evaluate
3.	Experiment with equipment's to find out various environmental parameters.	L3 - Apply
4.	Utilize the personal protective equipment in-dependently.	L3 - Apply
5.	Identify the various problems with the use of software and hence to predict the real situations on major accidents.	L3 - Apply

TOTAL(P): 30 Periods



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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1		3				2
CO2			3		2	
CO3			2	3		
CO4		3			3	
CO5	2					2
Average	2	3	2.5	3	2.5	2
1-Low, 2 -Medium, 3-High.						



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UX23PT801	TECHNICAL SEMINAR / CASE STUDY PRESENTATION	CP	L	T	P	C
		2	0	0	2	NC
Programme & Branch	Common to all M.E (SE, AE, PED, ISE) and M.C.A.	Version : 1.0				
Course Objectives:						
1	To encourage the students to study advanced engineering developments.					
2	To prepare and present the technical and case study reports.					
Method of Evaluation:						
The students need to identify an area of interest or topic in their programme of study or case study and prepare a 5-10 page report and a presentation. Based on the report and presentation, the course is evaluated for 100 marks. Minimum 50 marks is essential to pass. In case a student fails, he has to make such presentation in the subsequent semesters. The evaluation guidelines will be issued by the Head of the Department before the commencements of the course. The objectives are improving literature searching capabilities, comprehension and ability to write reports and to make presentations. It is assessed in Internal Assessment mode only and no End Semester Examination.						
						Total(P) : 30 Periods
Course Outcomes:						BLOOM'S Taxonomy
At the end of this course, the students will demonstrate the ability to						
CO1	Perform the review and present technological developments in their field.					L3 - Apply
CO2	Interpret the case study report and make a decision.					L3 - Apply

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1	3	1	1	1	2
CO2	1	3	1	2	1	1
Average	1	3	1	1.5	1	1.5
1-Low, 2 -Medium, 3-High.						

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ME23IS309	ELECTRICAL SAFETY	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To provide knowledge on basics of electrical fire and statutory requirements for electrical safety.					
2	To understand the causes of accidents due to electrical hazards.					
3	To learn the various protection systems in Industries from electrical hazards.					
4	To learn the importance of earthing.					
5	To distinguish the various hazardous zones and applicable fire proof electrical devices.					
UNIT-I	CONCEPTS AND STATUTORY REQUIREMENTS	9				
Introduction – electrostatics (L1), electro magnetism(L1), stored energy(L1), energy radiation and electromagnetic interference(L1) – Working principles of electrical equipment(L2)-Indian electricity act and rules(L1)-statutory requirements from electrical inspectorate(L1)-international standards on electrical safety (L1)– first aid-cardio pulmonary resuscitation(CPR) (L1).						
UNIT-II	ELECTRICAL HAZARDS	9				
Primary and secondary hazards (L2)-shocks(L1), burns(L1), scalds(L1), falls-human safety in the use of electricity(L1). Energy leakage(L2)-clearances and insulation(L2)-classes of insulation(L2)-voltage classifications(L2)-excess energy current surges(L2)-Safety in handling of war equipment's (L2)-over current and short circuit current(L2)-heating effects of current(L2)-electromagnetic forces(L1)-corona effect(L2)-static electricity(L1) –definition, sources, hazardous conditions(L2), control(L2), electrical causes of fire and explosion(L2)-ionization, spark and arc ignition energy(L2)-national electrical safety code ANSI(L2).Lightning (L2), hazards (L2), lightning arrestor (L2), installation – earthing(L2), specifications(L2), earth resistance(L2), earth pit maintenance(L2).						
UNIT-III	PROTECTION SYSTEMS	9				
Fuse(L1), circuit breakers and overload relays(L1) – protection against over voltage and under voltage (L2)– safe limits of amperage – voltage –safe distance from lines(L2)-capacity and protection of conductor-joints-and connections(L2), overload and short circuit protection(L2)-no load protection(L2)-earth fault protection(L2). FRLS insulation(L2)-insulation and continuity test(L2)-system grounding(L2)-equipment grounding(L2)-earth leakage circuit breaker (ELCB) (L2)-cable wires(L2)-maintenance of ground-ground fault circuit interrupter(L2)-use of low voltage(L2)-electrical guards(L2)-Personal protective equipment(L2) – safety in handling hand held electrical appliances tools and medical equipment's (L2).						


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UNIT-IV	SELECTION,INSTALLATION,OPERATION AND MAINTENANCE	9
Role of environment in selection(L2)-safety aspects in application(L2) - protection and interlock(L2)-self diagnostic features and fail safe concepts(L2)-lock out and work permit system(L2)-discharge rod and earthing devices safety in the use of portable tools(L2)-cabling and cable joints(L2)-preventive maintenance(L2).		
UNIT-V	HAZARDOUS ZONES	9
Classification of hazardous zones(L2)-intrinsically safe and explosion proof electrical apparatus(L2)-increase safe equipment(L2)-their selection for different zones(L2)-temperature classification(L2)-grouping of gases(L2)-use of barriers and isolators(L2)-equipment certifying agencies(L2).		
Total:45 Periods		
OPEN ENDED PROBLEMS /QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination		
Course outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Summarize the basic concepts in electrical circuit and its operations.	L2-Understand
CO2	Outline the electrical hazards in an Industry.	L2-Understand
CO3	Choose various protection systems for different electrical operations.	L3- Apply
CO4	Apply knowledge for the safe selection, installation, operation, and maintenance of electrical systems.	L3- Apply
CO5	Identify hazardous zones in various industries.	L3- Apply
REFERENCE BOOKS:		
1.	"Accident prevention manual for industrial operations", N.S.C. Chicago, 1982.	
2.	"Indian Electricity Act and Rules", Government of India, 2003.	
3.	"Power Engineers Handbook", TNEB, Chennai, 1989.	
4.	Martin Glov, "Electrostatic Hazards in Powder Handling", Research Studies Pvt. Ltd., England, 1988.	
5.	Fordham Cooper, W., "Electrical Safety Engineering", Butterworth and Heinemann Ltd, London, Third edition, 1998.	
VIDEO REFERENCES:		
1.	https://www.youtube.com/watch?v=tt800iM1N9s	
2.	https://www.youtube.com/watch?v=MEk68_veQYM	

WEB REFERENCES:

1. <https://www.osha.gov/electrical#:~:text=Electricity%20has%20long%20been%20recognized,electrocution%2C%20fires%2C%20and%20explosions.>
2. <https://www.ncbi.nlm.nih.gov/books/NBK580528/>

ONLINE COURSES:

1. https://onlinecourses.swayam2.ac.in/nou20_cs08/preview
2. <https://www.tcsion.com/courses/ve/safety/siemens/electrical-safety-online-course-and-training/>

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1		2	3			2
CO2		2	3			
CO3		2	3	1		
CO4	1	2	3	1		1
CO5	2	2	3	1	1	1
Average	1.5	2	3	1	1	1.3
1-Low,2-Medium,3-High.						

Beyond Knowledge

ME23IS601	PROJECT WORK-PHASE I	CP	L	T	P	C
		12	0	0	12	6
Programme & Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To identify relevant research problems by searching academic databases and literature.					
2	To design and conduct preliminary studies to explore identified problems.					
3	To compile and present research findings effectively.					
COURSE CONTENT:						
The Student will identify and select a problem based on comprehensive literature survey. The student should submit a proposal and get it approved by the Head of the department.						
Three reviews will be conducted by Project review committee. Students will be evaluated by the committee during the review and suggestions will be offered by members.						
The report for PHASE-I should be submitted by the students at the end of course						
						Total: 180 Periods
Course Outcomes:						BLOOM'S Taxonomy
Upon completion of this course the students will be able to:						
CO1	Collect various literature articles to identify a research gap related to Environmental Health and Safety (EHS).					L3- Apply
CO2	Analyze the problem using suitable methodology to give solution					L4- Analyze
CO3	Prepare a project report and present the findings.					L3- Apply

Beyond Knowledge

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	3	3	1	1	1
CO2	3	3	3	2	2	1
CO3	1	3	2	1	1	1
Average	2	3	2.7	1.3	1.3	1
1-Low, 2-Medium, 3-High.						

ME23IS602	PROJECT WORK–PHASE II	CP	L	T	P	C
		24	0	0	24	12
Programme& Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To develop the skill of students for analyzing safety problems to control the hazard.					
2	To expose the students to identify and evaluate the hazards in an industry under study.					
3	To expose the students to assess the Compliance level of safety norms and procedures.					
COURSE CONTENT:						
It is the continuation of Phase I project. Three reviews will be conducted by Project review committee. Students’ presentation will be evaluated by the committee during the reviews and suggestions will be offered by the committee members.						
At least one paper should be published by the student in international/national conference and/or filing of the patent.						
There port should be submitted by the students at the end of course.						
Total: 360 Periods						
Course Outcomes:						
Upon completion of this course the students will be able to:						BLOOM’S Taxonomy
CO1	Apply the appropriate methodology and find the solution for EHS related problems.					L4- Analyze
CO2	Create Safety procedures, and accident investigation reports relevant to the problem identified.					L6- Create
CO3	Prepare a project report and present the findings.					L3- Apply

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	2	3	3	2	1
CO2	3	3	3	3	3	1
CO3	1	3	2	1	1	1
Average	2.3	2.7	2.7	2.3	2	1
1-Low,2-Medium,3-High.						

ME23IS410	SAFETY IN TEXTILE INDUSTRY	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To provide the student about the basic knowledge about the textile industries and its products by using various machineries.					
2	To enforce the knowledge on textile processing and various processes in making the yarn from cotton or synthetic fibers.					
3	To understand the various hazards of processing textile fibers by using various activities.					
4	To inculcate the knowledge on health and welfare activities specific to the Textile industries as per the Factories Act.					
5	To provide the student about the basic knowledge about the textile industries and its products by using various machineries.					
UNIT-I	INTRODUCTION	9				
Introduction to process flow charts of i) short staple spinning, ii) long staple spinning, iii) viscose rayon and synthetic fiber, manufacturer, iv) spun and filament yarn to fabric manufacture, v) jute spinning and jute fabric manufacture-accident hazard, (L2)-guarding of machinery and safety precautions in opening,(L3)- carding, combing, drawing, flyer frames and ring frames, doubles, rotor spinning,(L2) winding, warping, softening/spinning specific to jute(L2).						
UNIT-II	TEXTILE HAZARDS I	9				
Accident hazards L2i)sizing processes-cooking vessels, transports of size, hazards due to steam(L3) ii)Loom shed-shuttle looms and shuttles looms(L2) iii)knitting machines iv)non-wovens(L2).						
UNIT- III	TEXTILE HAZARDS II	9				
Scouring, bleaching(L2),dyeing, punting, mechanical finishing operations(L2), and effluents in textile processes (L2).						
UNIT-IV	HEALTH AND WELFARE	9				
Health hazards in textile industry related to dust (L2), fly and noise generated-control measures-relevant occupational diseases (L2) personal protective equipment(L2)-health and welfare measures specific to textile industry(L2), Special precautions for specific hazardous work environments(L2).						
UNIT-V	SAFETY STATUS	9				
Relevant provision of factories act and rules and other statues applicable to textile industry (L2) - effluent treatment and waste disposal in textile industry(L3).						
Total: 45 Periods						
OPEN ENDED PROBLEMS / QUESTIONS						
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as Internal Assessment only and not for the End Semester Examination.						

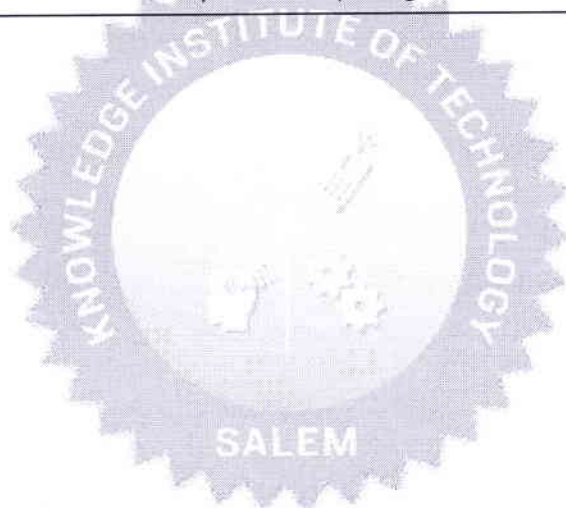
Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Identify potential accident hazards associated with various stages of textile manufacturing processes.	L3- Apply
CO2	Apply safety precautions in loom operations.	L3- Apply
CO3	Summarize specific accident hazards present in loom shed environments, including those associated with both shuttle and non-shuttle looms.	L2-Understand
CO4	Apply control measures to reduce health hazards in the textile industry and implement appropriate personal protective equipment (PPE) to ensure worker safety.	L3- Apply
CO5	Apply knowledge of waste disposal in the textile industry, including solid waste, hazardous waste, and wastewater sludge, in accordance with regulatory requirements.	L3- Apply
REFERENCE BOOKS:		
1.	"100 Textile fires analysis, findings and recommendations" LPA, 2008.	
2.	Groover E.B. and Hamby D.S., "Hand book of textile testing and quality control", New York: Textile Book Publishers, 1960.	
3.	"Quality tolerances for water for textile industry", BIS, 1992.	
4.	Shenai V.A., "A technology of textile processing", Vol. I, Textile Fibres, Sevak, 1975.	
5.	Little A.H., Water supplies and the treatment and disposal of effluent, Manchester: Textile Institute, 1975.	
6.	"Safety in Textile Industry", Thane Belapur Industries Association, Mumbai, 2007.	
VIDEO REFERENCES:		
1.	https://www.youtube.com/watch?v=j-XNzBUKOoE	
2.	https://www.youtube.com/watch?v=XADuwFDOyz0&pp=ygUPaGF6YXJkIGFuYWx5c2lz	
WEB REFERENCES:		
1.	https://www.graphicproducts.com/articles/hazard-analysis-risk-assessment/	
2.	https://www.aiche.org/ccps/introduction-hazard-identification-and-risk-analysis	
ONLINE COURSES:		
1.	https://onlinecourses.nptel.ac.in/noc23_mg98/preview?user_email=tdmech@kiot.ac.in	
2.	https://onlinecourses.swayam2.ac.in/nou23_ge81/preview	

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1		1	3		2	
CO2		2	3	3		2
CO3		1	3			
CO4	1	2	3	2		
CO5		1	2		2	
Average	1	1.4	2.8	2.5	2	2
1-Low, 2-Medium, 3-High.						

ME23IS411		SAFETY IN MINES				CP	L	T	P	C
						3	3	0	0	3
Programme & Branch		M.E.INDUSTRIAL SAFETY ENGINEERING				Version:1.0				
Course Objectives:										
1	To provide in depth knowledge on Safety of mines of various types.									
2	To understand the different types of mines and risks involved in the mining operations.									
3	To gain knowledge on types of accidents in mines and how to manage during accidents.									
4	To assess the Hazardous nature of mining activities and develop a safety system to reduce the risk									
5	To implement the Emergency preparedness in the working environment of mines and to plan for the disaster management.									
UNIT-I		OPEN CASTMINES				9				
Causes and prevention of accident from : Heavy machinery, belt and bucket conveyors, drilling, hand tools(L2)-pneumatic systems, pumping, water, dust, electrical systems, fire prevention(L2). Garage safety(L2)-accident reporting system(L2)-working condition(L2)-safe Transportation(L2)-handling of explosives(L2).										
UNIT-II		UNDER GROUND MINES				9				
Fall of roof and sides-effect of gases(L2)- fire and explosions-water flooding(L2)-warning sensors(L2)-gas detectors- occupational hazards(L2)-working conditions(L2)-winding and transportation(L2).										
UNIT-III		TUNNELLING				9				
Hazards from: ground collapse(L2), inundation and collapse of tunnel face, falls from platforms and danger from falling bodies(L2). Atmospheric pollution (gases and dusts) (L2)- trapping (L2)-transport(L2)-noise(L2)- electrical hazards-noise and vibration from: pneumatic tools and other machines(L2)-ventilation and lighting(L2)-personal protective equipment(L2).										
UNIT-IV		RISK ASSESSMENT				9				
Basic concepts of risk(L2)-reliability and hazard potential(L2)-elements of risk assessment(L2) - statistical methods (L3)- control charts(L2)-appraisal of advanced techniques(L3) - fault tree analysis(L5) - failure mode and effect analysis(L5)-quantitative structure(L2) - activity relationship analysis(L2) - fuzzy model for risk assessment(L2).										
UNIT-V		ACCIDENT ANALYSIS AND MANAGEMENT				9				
Accidents classification and analysis(L2)-fatal(L2), serious, minor and reportable accidents (L2)- safety audits(L3)- recent development of safety engineering approaches for mines(L2) - frequency rates(L3) - accident occurrence(L2) - investigation -measures for improving safety in mines(L2) - cost of Accident(L3) - emergency preparedness(L2)-disaster management(L2).										
Total:45 Periods										

OPEN ENDED PROBLEMS/QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination		
Course outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Interpret the concept of safety aspects in the mining industries.	L2-Understand
CO2	Summarize the hazards and control measure in an underground mining activity.	L2-Understand
CO3	Apply control measures to address the hazards encountered in tunneling activities to ensure worker safety	L3- Apply
CO4	Assess the severity of risk in mines to take the required remedial action.	L5-Evaluate
CO5	Utilize risk assessment techniques, disaster management, and emergency preparedness to prevent accidents.	L3- Apply
REFERENCE BOOKS:		
1.	"DGMS Circulars-Ministry of Labour", Government of India press, OR Lovely Prakashan-DHANBAD, 2002.	
2.	Kejiriwal B.K. "Safety in Mines", Gyan Prakashan, Dhanbad,2001.	
3.	Michael Karmis "Mine Health and Safety Management",SME,Littleton,Co.2001.	
4.	Dhillon, Balbir S "Mine safety- A modern Approach", Springer Publication,2010.	
VIDEO REFERENCES:		
1.	https://www.youtube.com/watch?v=fEFZw7bXSmk&list=PLB3JRydr2LBWmZ0n54wDrJHqzlsaf4bF0	
2.	https://www.youtube.com/watch?v=VE_xMqMp0k&list=PL8sSTcOtMi6a5saSaUnpQjIFtjQ3qw2lt	
WEB REFERENCES:		
1.	https://www.dgms.gov.in/	
2.	https://coal.gov.in/sites/default/files/2020-09/Chapter11-en.pdf	
ONLINE COURSES:		
1.	https://onlinecourses.nptel.ac.in/noc22_mm47/preview	
2.	https://www.classcentral.com/course/mining-the-university-of-queensland-health-safety-22045	

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1	2	3			
CO2	1	2	3			
CO3	1	2	3	1		
CO4	2	2	3			1
CO5	2	2	3	1	1	1
Average	1.4	2	3	1	1	1
1-Low,2-Medium,3-High.						



Beyond Knowledge

ME23IS412	DOCK SAFETY	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To understand safety legislation related to dock activities in India.					
2	To understand the causes and effects of accidents during dock activities.					
3	To know the various material handling equipment and lifting appliances in dock.					
4	To know the safe working on board the ship and storage in the yards.					
5	To understand the safe operation of crane, portainers, lift trucks and container handling equipment.					
UNIT-I	HISTORY OF SAFETY LEGISLATION	9				
History of dock safety statues in India-background of present dock safety statues (L2)-dockworkers (safety, health and welfare) act 1986 (L2) and the rules and regulations framed there under, other statues like marking of heavy packages act 1951 (L2) and the rules framed there under (L2) -manufacture , storage and import of hazardous chemicals (L2). Rules 1989 framed under the environment (protection) act, (L2) 1989-fewcases laws to interpret the terms used in the dock safety statues(L2). Responsibility of different agencies for safety, health and welfare involved in dock work (L2) -responsibilities of port authorities – dock labour board (L2) – owner of ship master, agent of ship(L2)-owner of lifting appliances and loose gear etc. (L2) – employers of dock workers like stevedores (L2) –clearing and forwarding agents (L2) – competent persons and dock worker. Forums for promoting safety and health in ports (L2)-Safe Committees and Advisory Committees. Their functions, training of dock workers(L2).						
UNIT-II	WORKING ON BOARD THE SHIP	9				
Types of cargo ships (L2) – working on board ships(L2) – Safety in handling of hatch beams (L2) – hatch covers including its marking, Mechanical operated hatch covers of different types and its safety features (L2) – safety in chipping and painting operations on board ships (L2) – safe means of accesses – safety in storage etc. (L2) – illumination of decks and in holds (L2) – hazards in working inside the hold of the ship and on decks – safety precautions needed (L2) – safety in use of transport equipment (L2) -internal combustibile engines like fort-lift trucks-pay loaders, etc (L2). Working with electricity and electrical management (L2) – Storage – types, hazardous cargo(L2).						

UNIT-III	LIFTING APPLIANCES	9
Different types of lifting appliances (L2)- construction, maintenance and use, various methods of rigging of derricks(L2), safety in the use of container handling/lifting appliances like portainers, transtainer, top lift trucks and other containers (L2)-testing and examination of lifting appliances(L2) – portainers – transtainer stop lift trucks(L2)- derricks in different rigging etc(L2). Use and care of synthetic and natural fiber ropes (L2) – wire rope chains, different types of slings and loose gears(L2).		
UNIT- IV	TRANSPORT EQUIPMENT	9
The different types of equipment for transporting containers and safety in their use-safety in the use of self loading container vehicles(L2), container side lifter, fork lift truck, dock railways, conveyors and cranes(L2). Safe use of special lift trucks inside containers(L2) – Testing, examination and inspection of containers(L2) – carriage of dangerous goods in containers and maintenance and certification of containers for safe operation Handling of different types of cargo (L2) – stacking and unstacking both on board the ship and a shore(L2) – loading and unloading of cargo identification of berths/walking for transfer operation of specific chemical from ship to shore and vice versa (L2) – restriction of loading and unloading operations(L3).		
UNIT-V	EMERGENCY ACTION PLAN AND DOCK WORKERS(SHW) REGULATIONS 1990	9
Emergency action Plans for fire and explosions (L2) - collapse of lifting appliances and buildings, sheds etc., (L3) - gas leakages and precautions concerning spillage of dangerous goods etc(L2)., -Preparation of on-site emergency plan and safety report(L2). Dock workers(SHW) rules and regulations1990- related to lifting appliances(L2), Container handling, loading and unloading (L2), handling of hatch coverings and beams, Cargo handling, conveyors, dock railways, forklift(L2).		
Total:45 Periods		
OPEN ENDED PROBLEMS / QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.		

Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
C01	Illustrate the background and evolution of present dock safety statutes, including the factors that contributed to their enactment.	L2-Understand
C02	Summarize the importance of safety measures when handling hatch beams and hatch covers on cargo ships.	L2-Understand
C03	Outline the principles of safe rigging practices, including the selection and placement of rigging equipment and the calculation of load capacities.	L2-Understand
C04	Select safety measures to prevent accidents and ensure the safe handling of containers and cargo during transportation operations."	L3- Apply
C05	Make use of protocols and procedures for responding to emergencies effectively, including evacuation plans and emergency response teams.	L3- Apply
REFERENCE BOOKS:		
1.	"Dock Safety" Thane Belapur Industries Association, Mumbai.	
2.	Bindra S R "Course in Dock and Harbour Engineering", Dhanpat Rai Publications (P) Ltd., New Delhi, 2013.	
3.	"Safety and Health in Dock work", 2 nd Edition, ILO,1992.	
4.	Srinivasan "Harbour, Dock and Tunnel Engineering", Charotar Publishing House Pvt. Limited, 2011.	
5.	Taylor D.A., "Introduction to Marine Engineering", 2 nd Edition, Butterworth-Heinemann, 1996.	
VIDEO REFERENCES:		
1.	https://www.youtube.com/watch?v=b9cTL5JakVc	
2.	https://www.youtube.com/channel/UC7j-KnWLS8at_Z0c0ZbunoA	
WEB REFERENCES:		
1.	https://www.ehs.uci.edu/safety/_pdf/loading-dock-safety-reference-guide.pdf	
2.	https://dgfasli.gov.in/dock-safety-view	
ONLINE COURSES:		
1.	https://nptel.ac.in/courses/114105003	
2.	https://archive.nptel.ac.in/courses/114/105/114105003/	

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
C01		1	3		3	
C02		1	3	1		
C03		1	3	1		
C04	1	1	3	2	2	
C05		1	2			2
Average	1	1	2.8	0.8	1	0.4
1-Low, 2-Medium, 3-High.						



Beyond Knowledge

ME23IS413	SAFETY IN ENGINEERING INDUSTRY	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1.	To know the safety rules and regulations, standards and codes.					
2.	To study various mechanical machines and their safety importance					
3.	To understand the principles of machine guarding and operation of protective devices.					
4.	To know the working principle of mechanical engineering processes such as metal forming and joining process and their safety risks.					
5.	To develop the knowledge related to health and welfare measures in engineering industry					
UNIT-I	SAFETY IN METAL WORKING MACHINERY AND WOOD WORKING MACHINES					9
General safety rules (L2), principles(L2), maintenance(L2), Inspections of turning machines(L2), boring machines(L2), milling machine(L2), planning machine and grinding machines(L2), CNC machines(L2), Wood working machinery (L2), types (L2),safety principles (L2), electrical guards, work area (L2),material handling (L2) ,inspection (L2),standards and codes (L2)-saws, types, hazards (L2).						
UNIT-II	PRINCIPLES OF MACHINE GUARDING					9
Guarding during maintenance (L2), Zero Mechanical State (ZMS) (L2), Definition, Policy for ZMS (L2)-guarding of hazards (L2)- point of operation protective devices(L2), machine guarding, types, fixed guard, interlock guard, automatic guard, trip guard, electron eye, positional control guard, fixed guard fencing(L2)-guard construction(L2)- guard opening(L2).						
Selection and suitability: lathe-drilling-boring-milling-grinding-shaping-sawing-shearing-presses (L3)-forgehammer-flywheels-shafts-couplings-gears-sprocketswheelsandchains (L3)-pulleys and belts (L3)-authorized entry to hazardous installations (L3)-benefits of good guarding systems (L2).						
UNIT-III	SAFETY IN WELDING AND GAS CUTTING					9
Gas welding and oxygen cutting(L2), resistances welding, arc welding and cutting, common hazards , personal protective equipment, training, safety precautions in brazing, soldering and metalizing(L2)-explosive welding, selection, care and maintenance of the associated equipment and instruments(L3)- safety in generation, distribution and handling of industrial gases(L2)- colour coding(L2)-flash back arrestor(L2)-leak detection(L2)-pipe line safety(L2)-storage and handling of gas cylinders(L2).						

UNIT-IV	SAFETY IN COLD FORMING AND HOT WORKING OF METALS	9
Cold working (L2), power presses (L2), point of operation safe guarding (L2), auxiliary mechanisms (L2), feeding and cutting mechanism (L2), hand or foot-operated presses (L2), power press electric controls (L2), power press set up and die removal (L2), inspection and maintenance (L2) -metal sheers-press brakes(L2). Hot working safety in forging (L2), hot rolling mill operation (L2), safe guards in hot rolling mills (L2) -hot bending of pipes (L2), hazards and control measures(L2) . Safe tying as furnace operation, cupola, crucibles, ovens (L2), foundry health hazards (L2), work environment(L2), material handling in foundries(L2), foundry production cleaning and finishing foundry processes (L2).		
UNIT-V	SAFETY IN FINISHING, INSPECTION AND TESTING	9
Heat treatment operations (L2), electro plating (L2), paint shops(L2), sand and shot blasting(L2), safety in inspection and testing(L2), dynamic balancing(L2), hydro testing(L2), valves, boiler drums and headers(L2), pressure vessels(L2), air leak test(L2), steam testing(L2), safety in radiography(L2), personal monitoring devices(L2), radiation hazards(L2), engineering and administrative controls(L2), Indian Boilers Regulation(L2). Health and welfare measures in engineering industry (L2)-pollution control in engineering industry (L2)-industrial waste disposal (L2).		
Total:45 Periods		
OPEN ENDED PROBLEMS/QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination		
Course outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Infer safety rules, standards and codes in various mechanical engineering processes.	L2-Understand
CO2	Choose suitable machine guarding systems for various machines such as lathes, drilling machines, boring machines, milling machines, etc.	L3 - Apply
CO3	Apply safety concepts to welding, gas cutting, storage and handling of gas cylinders, and metal forming processes.	L3 - Apply
CO4	Demonstrate knowledge in testing and inspection as per rules in boilers, heat treatment operations.	L2-Understand
CO5	Apply preventive measures to enhance the health and welfare of workers	L3 - Apply

REFERENCE BOOKS:

1.	Philip E. Hagan, John Franklin Montgomery, James T. O'Reilly, "Accident Prevention Manual" NSC, Chicago, 13th edition, 2009.
2.	"Occupational safety Manual" BHEL, Trichy, 1988.
3.	John V. Grimaldi and Rollin H. Simonds "Safety Management", All India Travelers Bookseller, New Delhi, 1989.
4.	Krishnan N.V. "Safety in Industry" Jaico Publishery House, 1996.
5.	"Indian Boiler acts and Regulations", Government of India.
6.	"Safe use of wood working machinery", HSE, UK, 2005.
7.	"Health and Safety in welding and Allied processes" Welding Institute, UK, High Tech. Publishing Ltd., London, 1989.

VIDEO REFERENCES:

1.	https://www.youtube.com/watch?v=p9tJtV-SDXY
2.	https://www.youtube.com/watch?v=bAPMLwi0a88

WEB REFERENCES:

1.	https://www.osha.gov/woodworking
2.	https://www.osha.gov/sites/default/files/publications/osh3157.pdf

ONLINE COURSES:

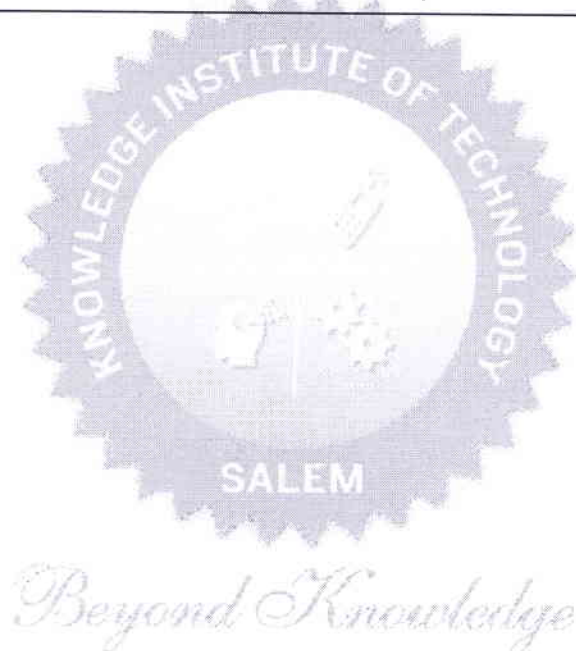
1.	https://www.aws.org/Certification-and-Education/Education/Safety-in-Welding/
2.	https://www.classcentral.com/subject/woodworking

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1		2	3		1	
CO2		2	3		1	1
CO3		2	3		1	
CO4		2	3		1	
CO5	2	2	3	2	1	
Average	2	2	3	2	1	1
1-Low, 2-Medium, 3-High.						

ME23IS414	QUALITY ENGINEERING IN PRODUCTION SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To understand safety legislation related to dock activities in India.					
2	To understand the causes and effects of accidents during dock activities.					
3	To learn the various material handling equipment and lifting appliances in dock.					
4	To know the safe working on board the ship and storage in the yards.					
5	To understand the safe operation of crane, portainers, lift trucks and container handling equipment					
UNIT-I	INTRODUCTION TO QUALITY ENGINEERING AND LOSS FUNCTION	9				
Quality value and engineering (L2) - overall quality system (L2) -quality engineering in product design (L2) -quality engineering in design of production processes (L2) - quality engineering in production (L2)-quality engineering in service. Loss function Derivation (L2) – use-loss function for products/system (L2)-justification of improvements- loss function and inspection(L2) -quality evaluations and tolerances -N type, S type, L type(L2).						
UNIT-II	ON-LINE QUALITY CONTROL	9				
On-line feedback quality control variable characteristics-control with measurement interval (L2) - one unit, multiple units-control systems for lot and batch production(L2). On-line process parameter control variable characteristics (L2)- process parameter tolerances(L2)-feedback control systems- measurement error and process control parameters(L2).						
UNIT-III	ON-LINE QUALITYCONTROL ATTRIBUTES AND METHODS FOR PROCESS IMPROVEMENTS	9				
Checking intervals (L2)- frequency of process diagnosis(L3). Production process improvement method (L2)- process diagnosis improvement method(L2)-process adjustment and recovery improvement methods(L2).						
UNIT- IV	QUALITY ENGINEERING AND TPM	9				
Preventive maintenance schedules(L2) -PM schedules for functional characteristics(L3) -PM schedules for large scale systems. Quality tools(L2)– fault tree analysis, event tree analysis, (L2)failure mode and effect analysis. (L2) ISO quality systems(L2).						
UNIT-V	SIX SIGMA AND ITS IMPLEMENTATION	9				
Introduction- definition-methodology (L2)- impact of implementation of six sigma-DMAIC method (L2)- roles and responsibilities-leaders ,champion, black belt, greenbelts (L2). Do's and don'ts - readiness of organization (L2)– planning - management role - six sigma tools(L2) – sustaining six sigma(L2).						
TOTAL:45 Periods						

OPEN ENDED PROBLEMS / QUESTIONS		
Course specific open ended problems will be solved during the class room teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.		
Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Outline the role of quality engineering in the design phase of products, emphasizing the integration of quality considerations from the outset.	L2-Understand
CO2	Summarize the role of process parameters in influencing product quality and explain how to adjust them in real-time to maintain quality standards.	L2-Understand
CO3	Utilize various methods and techniques to improve production processes, enhancing both product quality and efficiency.	L3- Apply
CO4	Apply fault tree analysis, event tree analysis, and failure mode and effect analysis(FMEA) to identify potential failure modes.	L3- Apply
CO5	Utilize strategic planning and change management to effectively prepare an organization for Six Sigma implementation.	L2-Understand
REFERENCE BOOKS:		
1.	Brue G, "Six Sigma or Managers", Tata-McGrawHill, New Delhi, Second reprint,2002.	
2.	De Feo J A and Barnard W W, "Six Sigma:Breaktrough and Beyond",Tata McGraw-Hill, New Delhi,2005.	
3.	Pyzdek T and Berger R W," Quality Engineering Handbook",Tata-McGrawHill, New Delhi, 1996.	
4.	Taguchi G, Elsayed E A and Hsiang, T.C.,"Quality Engineering in Production Systems", Mc-Graw-Hill Book company, Singapore, International Edition,1989.	
VIDEO REFERENCES:		
1.	https://www.youtube.com/watch?v=SoUjQpIO3YY&list=PLeGWvtOKhUv78mHlXeyPtqjhmaWZQNmyK	
2.	https://www.youtube.com/watch?v=H2z4pi0KZSs&list=PLeGWvtOKhUv78mHlXeyPtqjhmaWZQNmyK&index=4 .	
WEB REFERENCES:		
1.	https://quality-one.com/qualityengineering/#:~:text=Quality%20Engineering%20consists%20of%20analysis,the%20customer's%20requirements%20and%20expectations.	
2.	http://www1.iitkgp.ac.in/downloads/sm_gian_1617_qe.pdf	
ONLINE COURSES:		
1.	https://onlinecourses.nptel.ac.in/noc20_mg18/preview	
2.	https://nptel.ac.in/courses/110105088	

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
C01		1	2			
C02		1	2			
C03		1	2		1	
C04	1	2	3		1	1
C05	2	2	3	2	1	
Average	1.5	1.4	2.4	2	1	1
1-Low,2-Medium,3-High.						



ME23IS415	ISO45001 AND ISO14000	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	The course could provide the basic knowledge on Occupational Health and Safety Management System and Environmental Management System standards.					
2	To inculcate the knowledge on various terms and terminologies which are used in the Occupational Health, Safety and Environmental Management system.					
3	To inculcate the knowledge on performance evaluation methods based on ISO 45001.					
4	To educate about the various steps to be taken for certification of ISO45001 (Occupational health and safety management systems) and ISO14001 (Environmental Management Systems) standards.					
5	To impart knowledge on Environment Impact Assessment (EIA), Life Cycle Assessment of product and principles of Eco labeling.					
UNIT-I	OH & S MANAGEMENT SYSTEM STANDARD	9				
Introduction to ISO 45001 (L1)–Development of various OHSMS standards (L2)– aim of OH& S management system(L2)–success factors(L2)– plan do check act cycle(L2)- contents and scope of ISO45001(L2)- terms and definitions(L2) –leadership and worker participation (L2)– leadership and commitment(L2) -OH & S policy(L2)- organizational roles and responsibilities and authorities(L2) –consultation and participation of workers(L2).						
UNIT-II	PLANNING	9				
Actions to address risk and opportunities (L2) – hazard identification and assessment of risks and opportunities (L2) –determination of legal and other requirements(L2)-planning action(L2)-OH & S objectives and planning to achieve them(L2)-support (L2)– resources (L2)–competence (L2) –awareness (L2) –communication (L2) –documented information (L2) – operation planning & control(L2)–management of change(L2)–procurement(L2)– Emergency preparedness and response(L2).						
UNIT-III	PERFORMANCE EVALUATION	9				
Monitoring, measurement, analysis and performance evaluation(L3)-evaluation of compliance(L3)- Internal audit-management review(L2)- Improvement(L2)- Incident, non conformity and corrective action(L2)–continual improvement-guidance of the use of the document ISO 45001(L2)-benefits of certification(L2)-certification procedure(L2).						
UNIT-IV	ISO14001 & ISO19011	9				
EMS, ISO14001, specifications, objectives, Environmental Policy, Guidelines and Principles (ISO 14004) (L2), clauses 4.1 to 4.5. Documentation requirements (L2), 3 levels of documentation for a ISO14000 based EMS (L2), steps in ISO 14001 (L2).						

Implementation plan (L2),Registration (L2),Importance of ISO 14000 to the Management (L2).Auditing ISO14000(L3)- General principles of Environmental Audit(L2), Auditor(L2), steps in audit(L2),Audit plan(L2).

ISO 19011(L2)- Guidelines for auditing management Systems (L2) -General principles, managing audit programme (L2) -audit activities (L3),steps in audit (L2),audit plan(L3)- competence of auditors(L2).

UNIT-V	ENVIRONMENT IMPACT ASSESSMENT	9
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ISO 14040(LCA), General principles of LCA, Stages of LCA, Report and Review(L2). ISO14020 (Eco labeling) – History, 14021, 14024, Type I labels, Type II labels, ISO 14024,principles, rules for eco labeling before company attempts for it (L2). Advantages. EIA in EMS, Types of EIA,EIA methodology EIS, Scope, Benefits(L2) .Audit- methodology (L2), Auditors Audit results management review (L2)-Continual improvement(L2).

Total:45 Periods

OPEN ENDED PROBLEMS/QUESTIONS

Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination

Course outcomes:

Upon completion of this course the students will be able to:

BLOOM'S Taxonomy

CO1	Outline the various standards designed to maintain employee health and ensure environmental protection.	L2-Understand
CO2	Interpret the basic differences between the ISO 9000 series, ISO 45001, and ISO 14000 standards, including the various clauses that govern the maintenance of each standard.	L2-Understand
CO3	Apply various clauses of ISO 45001 and ISO 14000 to prepare procedures and related documents.	L3- Apply
CO4	Apply knowledge to prepare an ISO manual for obtaining certification from external certifying agencies.	L3- Apply
CO5	Select appropriate standards and clauses based on their relevance to various organizational types	L3- Apply

REFERENCE BOOKS:

1.	ISO 45001: 2018–Occupational Health and safety management systems requirements with guidance for use.
2.	ISO 14001:2004, Environmental Management Systems Requirements with Guidance for Use”, ISO, 2004.
3.	“Guidelines on Occupational Health and safety Management Systems (OSH-MS)” International Labour Organization,2001.

4.	"BS8800:2004 Occupational Health and Safety Management Systems Guide" BSI, UK, 2004.
5.	"ISO 19011:2011 Guidelines for Auditing Management Systems", ISO,2011.
6.	"ISO 14040:2006 Environmental Management- Life Cycle Assessment -Principles and Framework " ISO,2006.
7.	"ISO 14025:2006 Environmental Labels and Declarations -Type III Environmental Declarations - Principles and Procedures",ISO,2006.
8.	"ISO 14021:1999 Environmental Labels and Declarations-Self-Declared Environmental Claims (Type II environmental labeling)",ISO, 1999.
9.	"ISO 14020:2000 Environmental Labels and Declarations-General Principles", ISO, 2000.
VIDEO REFERENCES:	
1.	https://www.youtube.com/watch?v=7utZfepa140&list=PLO8v5ESo95YMIWanVt2nw1S8ZhzlAchs-
2.	https://www.youtube.com/watch?v=_3dNjtuqbmY
WEB REFERENCES:	
1.	https://www.iso.org/iso-45001-occupational-health-and-safety.html
2.	https://www.iso.org/standards/popular/iso-14000-family
ONLINE COURSES:	
1.	https://www.bsigroup.com/en-IN/occupational-health-and-safety-iso-45001/iso-45001-training-courses/
2.	https://www.bsigroup.com/en-IN/ISO-14001-Environmental-Management/Training-courses-for-ISO-14001/

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1		2	3		2	1
CO2		2	3		2	
CO3		2	3		2	
CO4		2	3		2	
CO5		2	3	1	2	
Average		2	3	1	2	1
1-Low,2-Medium,3-High.						

ME23IS416	ARTIFICIAL INTELLIGENCE AND DATA ANALYTICS	CP	L	T	P	C
		3	3	0	0	3
Programme& Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To understand data science fundamentals and its safety applications.					
2	To learn data acquisition, cleaning, and exploratory analysis for safety.					
3	To apply statistical concepts for safety evaluation and monitoring.					
4	To explore machine learning algorithms for safety event prediction and anomaly detection.					
5	To utilize visualization and communication techniques for safety data insights.					
UNIT-I	DATA SCIENCE AND SAFETY	9				
Introduction-Terminology in data science (L2) -.Application of data science (L2)-Overview of data science and its applications in safety-risk analysis (L2), anomaly detection and safety optimization (L2)-Introduction to safety analysis and risk assessment (L2).						
UNIT-II	DATA COLLECTION AND PREPROCESSING	9				
Data acquisition methods and sources relevant to Safety (L2)-Operators - Variables - data types -Conditional statements-Looping (L2) - Function -. Data structure (L2)- Lists, Dictionary and Tuple -Regular Expression (L2)-File Reading (CSV, Excel etc.) (L2)-Basics Class and Objects concepts (L2)-data cleaning handling missing values(L2)-dealing without liers-Exploratory data analysis techniques(L2).						
UNIT-III	STATISTICAL ANALYSIS	9				
Probability theory and statistical concepts relevant to safety analysis (L2) -statistical analysis, data visualization, and predictive modeling techniques(L2) - Descriptive and inferential statistics (L2) - Hypothesis testing and confidence intervals(L2)-Safety Program Evaluation, Comparative Analysis, Reliability Analysis(L2),Incident Investigation, Safety Performance Monitoring(L2).						
UNIT-IV	MACHINE LEARNING	9				
Introduction to supervised and unsupervised learning algorithms (L2)-Feature engineering and selection(L2)-Model evaluation and performance metrics (L2)-Regression models (L2)-Classification models(L2)-Ensemble methods and model optimization (L2)-Techniques for identifying anomalies and outlier detection(L2)-Time series analysis and forecasting for safety events (L2)-Event prediction using machine learning algorithms(L3).						

UNIT-V	VISUALIZATION AND COMMUNICATION OF SAFETY DATA	9
Data visualization techniques for safety insights - Story telling with data (L2) and effective communication of safety findings (L2)-Interactive dashboards and reporting tools for safety analysis(L2) - Ethical issues in working with safety data-Privacy and security concerns in data science for safety(L2)- Legal and regulatory frameworks related to safety data(L3)-Data science to safety problems(L3).		
Total:45 Periods		
OPEN ENDED PROBLEMS / QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.		
Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Interpret the benefits of using data science to improve safety measures and prevent accidents in various industries.	L2-Understand
CO2	Summarize the significant impact of data analysis on quality and effectiveness.	L2-Understand
CO3	Illustrate the understanding of safety-related data and draw meaningful insights from it.	L2-Understand
CO4	Identify trends, patterns, and correlations in experimental data.	L3- Apply
CO5	Organize data into a more understandable form, highlighting trends and outliers.	L3- Apply
REFERENCE BOOKS:		
1.	David J.Smith ,Kenneth GL Simpson "Safety Critical Systems Handbook: A Straight forward Guide to Functional Safety, IEC61508(2010 Edition) and Related Standards" 3rd Edition, Butterworth-Heinemann,2011.	
2.	Tim Kelly "Safety Critical Systems: Problems, Process and Practice". Springer London, 2009.	
3.	Nicholas J. Bahr "System Safety Engineering and Risk Assessment: A Practical Approach " 2nd edition, CRC Press, 2014.	
4.	Mariy Yao, Adelyn Zhou, and Marlene Jia "Applied Artificial Intelligence :A Handbook For Business Leaders" Topbot, 2018.	
5.	Charles D. Reese and James P. Nelson "Handbook of Safety and Health for the Service Industry" 1st edition, , CRC Press, 2008.	
6.	Jose L. Munoz and Luis F. Miranda-Moreno "Data Science for Transport: A Self-Study Guide with Computer Exercises" 1st edition, Springer International Publishing AG, 2018.	
7.	Avrim Blum, John Hopcroft, and Ravindran Kannan "Foundations of Data Science" Cambridge University Press, 2020.	

VIDEO REFERENCES:	
1.	https://www.youtube.com/watch?v=3K-vJIVMi5A
2.	https://www.youtube.com/watch?v=pKeVMlkFpRc&list=PLwdnzIV3ogoXaceHrrFVZCJkbm_JaSHcH
WEB REFERENCES:	
1.	https://dl.acm.org/doi/pdf/10.1145/3550473
2.	https://www.coursera.org/articles/data-analytics
ONLINE COURSES:	
1.	https://onlinecourses.nptel.ac.in/noc22_cs56/preview
2.	https://onlinecourses.swayam2.ac.in/nou23_ge81/preview

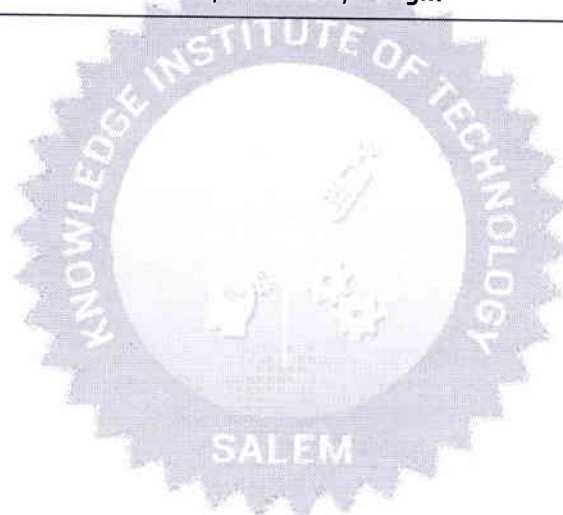
Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
C01				1		1
C02		1		2	2	
C03		1	2	2	1	
C04	1	1		2	1	
C05	1	1		2	1	
Average	1	1	2	1.8	1.3	1
1-Low,2-Medium,3-High.						

Beyond Knowledge

ME23IS417	DESIGN OF EXPERIMENTS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1.	To Impart knowledge on principles and steps in designing a statistically designed experiment.					
2.	To Build foundation in analyzing the data in single factor experiments and to perform post hoc tests.					
3.	To Provide knowledge on analyzing the data in factorial experiments.					
4.	Educate on analyzing the data analysis in special experimental designs and Response Surface Methods.					
5.	Impart knowledge in designing and analyzing the data in Taguchi's Design of Experiments to improve Process/Product quality.					
UNIT-I	EXPERIMENTAL DESIGN FUNDAMENTALS	9				
Importance of experiments(L1), experimental strategies(L2), basic principles of design, terminology(L2),ANOVA, steps in experimentation (L3), sample size, normal probability plot, linear regression models(L3).						
UNIT-II	SINGLE FACTOR EXPERIMENTS	9				
Completely randomized design (L3), Randomized block design(L3), Latin square design. (L3) Statistical analysis(L3), estimation of model parameters (L3), model adequacy checking (L3) pair wise comparison tests(L3).						
UNIT-III	MULTI FACTOR EXPERIMENTS	9				
Two and three factor full factorial experiments(L3), Randomized block factorial design(L3), Experiments with random factors (L3) ,rules for expected mean squares(L3),approximate F-tests, 2K factorial Experiments(L3).						
UNIT-IV	SPECIAL EXPERIMENTAL DESIGNS	9				
Blocking and confounding in 2K designs(L4). Two level Fractional factorial design(L4), nested designs, Split plot design(L4),Introduction to Response Surface Methods(L4).						
UNIT-V	TAGUCHI METHODS	9				
Steps in experimentation(L3), design using Orthogonal Arrays, data analysis(L3) ,Robust design- control and noise factors(L3), S/N ratios, parameter design(L3), Multi-level experiments, Multi- response optimization(L3), Introduction to Shainin DOE(L3).						
Total:45 Periods						

OPEN ENDED PROBLEMS / QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.		
Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Apply ANOVA to determine if there are significant differences among treatment means and find sources of variation in data.	L3- Apply
CO2	Identify model parameters in various experimental designs using appropriate statistical methods, such as least squares estimation.	L3- Apply
CO3	Plan and conduct experiments with random factors, understanding how randomization helps estimate treatment effects and improve the generalizability of results.	L3- Apply
CO4	Develop and analyze nested experiments with hierarchical structures, understanding their implications for experimental setup and statistical analysis.	L4- Analyze
CO5	Develop skills in multi-response optimization to simultaneously optimize multiple response variables and achieve optimal process settings.	L3- Apply
REFERENCE BOOKS:		
1.	Krishnaiah,K. and Shahabudeen,P. "Applied Design of Experiments and Taguchi Methods",PHI learning private Ltd.,2012.	
2.	Montgomery,D.C., "Design and Analysis of experiments", John Wiley and Sons, Eighth edition,2012.	
3.	Nicolo Belavendram, "Quality by Design; Taguchi techniques for industrial Experimentation", PrenticeHall,1995.	
4.	PhillipJ.Rose, "Taguchi techniques for quality engineering", McGraw Hill, 1996.	
5.	Montgomery,D.C., "Design and Analysis of Experiment, Minitab Manual", John Wiley and Sons, Seventh edition,2010.	
VIDEO REFERENCES:		
1.	https://www.youtube.com/watch?v=IEUTRhyoHNc&list=PLPjSqITyvDeWS9Lxp4jreGJ7eNsxHxJA8	
2.	https://www.youtube.com/watch?v=pKeVMlkFpRc&list=PLwdnzlV3ogoXaceHrrFVZCJKBm_laSHcH	
WEB REFERENCES:		
1.	https://home.iitk.ac.in/~shalab/anova/chapter4-anova-experimental-design-analysis.pdf	
2.	https://www.itl.nist.gov/div898/handbook/pmd/section3/pmd31.htm	
ONLINE COURSES:		
1.	https://onlinecourses.nptel.ac.in/noc21_mg48/preview	
2.	https://onlinecourses.swayam2.ac.in/aic23_ge17/preview	

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1	1				
CO2	2		1	1		
CO3			1	1		1
CO4			1	1	1	
CO5	2	1	1			
Average	1.7	1	1	1	1	1
1-Low,2-Medium,3-High.						



Beyond Knowledge

ME23IS418	RELIABILITY ENGINEERING	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	Impart knowledge in reliability concepts.					
2	Facilitate students in filling the life data into theoretical distribution.					
3	Educate the students in reliability evaluation of various configuration.					
4	Impart knowledge in reliability monitoring methods.					
5	Analyze effectively various techniques to improve reliability of the system.					
UNIT-I	RELIABILITY CONCEPTS	9				
Reliability definition(L2)-Quality and Reliability(L2) - Reliability mathematics(L2) - Reliability functions(L2)-Hazard rate(L2)-Measures of Reliability(L2)-Design life(L2)- A priori and posteriori probabilities(L2) - Mortality of a component(L2) - Mortality curve (L2)- Useful life(L2).						
UNIT-II	LIFE DATA ANALYSIS	9				
Data collection (L2) -Non Parametric methods: Ungrouped/Grouped, Complete/Censored data(L3)-Time to failure distributions: Exponential, Weibull-Probability plotting(L3)- Goodness of fit tests(L3).						
UNIT-III	PERFORMANCE EVALUATION	9				
Different configurations(L2) - Redundancy (L3)- k out of n system (L3)- Complex systems: RBD(L3) - Baye's approach(L3) - Cut and tie sets(L3)-Fault Trees(L3)-Stand by systems(L3).						
UNIT-IV	RELIABILITY MONITORING	9				
Life testing methods: Failure terminated (L3)- Time terminated (L3) - Sequential Testing (L3) - Reliability growth monitoring (L3) - Reliability allocation (L3) - Software reliability(L2)-Human reliability(L2).						
UNIT-V	RELIABILITY IMPROVEMENT	9				
Analysis of downtime - Repair time distribution - System repair time - Maintainability prediction-Measures of maintainability - Inspection decisions-System Availability.						
Total:45 Periods						

OPEN ENDED PROBLEMS / QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.		
Course outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Understand the basic concepts of reliability engineering.	L2-Understand
CO2	Apply the different technique stopper form life data analysis on a system.	L3- Apply
CO3	Apply the knowledge to conduct reliability assessment and failure analysis on any complex systems.	L3- Apply
CO4	Apply techniques to monitor reliability of the system.	L3- Apply
CO5	Analyze various techniques to improve reliability of the system.	L4- Analyze
REFERENCE BOOKS:		
1.	Charles E.Ebeling, "An introduction to Reliability and Maintain ability engineering", TMH,2000.	
2.	Roy Billington and Ronald N.Allan, "Reliability Evaluation of Engineering Systems", Springer,2007.	
VIDEO REFERENCES:		
1.	https://www.youtube.com/watch?v=BQXnKpP2lrI&t=15s	
2.	https://www.youtube.com/watch?v=uutg8jKrL9w&t=30s	
WEB REFERENCES:		
1.	https://reliably.com/blog/what-is-reliability-engineering/	
2.	https://study.com/academy/lesson/reliability-engineering-definition-purpose.html	
ONLINE COURSES:		
1.	https://onlinecourses.nptel.ac.in/noc23_ge20/preview	
2.	https://reliability-academy.com/	

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1		1	3		1	2
CO2	2	1	3		1	1
CO3	2	1	3			
CO4	1	1	3	2		
CO5	1	1	3		1	
Average	1.5	1	3	2	1	1.5
1-Low,2-Medium, 3-High.						

ME23IS419	LOGISTICS AND DISTRIBUTION MANAGEMENT	CP	L	T	P	C
		3	3	0	0	3
Programme& Branch	M.E.INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To provide knowledge on the fundamentals of logistics and warehouse safety.					
2	To understand the principles and practices of safe material handling and storage.					
3	To identify and mitigate risks associated with transportation safety.					
4	To implement safety protocols for dock and yard operations.					
5	To understand and apply fire safety measures in warehouses.					
UNIT-I	INTRODUCTION TO LOGISTICS					9
Introduction to logistics (L1) – Definition and significance (L1)– Role of logistics in supply chain management (L2) – Key logistics functions: transportation, warehousing, inventory management, order fulfillment (L2) – Safety regulations and standards in logistics (e.g., ISO 45001 for Occupational Health and Safety Management Systems) (L2) – Overview of safety management systems (L2) – Risk assessment in logistics operations (L2) – Emergency preparedness and response planning(L2).						
UNIT-II	MATERIAL HANDLING AND STORAGE					9
Principles of material handling (L2) – Types of material handling equipment (L1) – Safety in the use of conveyors, forklifts, cranes, and hoists (L2) – Ergonomics in material handling (L2) – Safe stacking and storage practices (L2) – Warehouse layout and design for safety (L3) – Palletizing and racking systems (L2) – Hazard identification and control in storage areas (L3) – Personal protective equipment (PPE) for material handling (L2) – Relevant standards: OSHA 1910 Subpart N (Materials Handling and Storage), ANSI/ITSDF B56.1 (Safety Standard for Low Lift and High Lift Trucks)(L2).						
UNIT-III	TRANSPORTATION SAFETY					9
Transportation modes and their safety considerations (L2) – Road transport safety: vehicle maintenance, driver training, and route planning (L2) – Safety in rail transport: loading/unloading, signaling systems, and emergency procedures (L2) – Air transport safety: cargo handling, security, and regulatory compliance (L2) – Maritime transport safety: ship loading/unloading, container handling, and port safety (L2) – Risk management in transportation (L5) – Incident investigation and reporting (L5) – Relevant standards: DOT regulations, IATA Dangerous Goods Regulations, IMDG Code (International Maritime Dangerous Goods) (L2).						

UNIT-IV	DOCK AND YARD SAFETY	9
Safety procedures for dock and yard operations (L1) – Dock design and layout for safety(L2) – Safe loading and unloading practices (L2) – Equipment used in dock and yard operations (L2) – Traffic management in yards (L2) – Worker safety and PPE requirements(L2) – Preventive maintenance of dock and yard equipment (L2) – Hazardous material handling and storage in docks (L2) – Emergency procedures for dock and yard incidents(L3) – Relevant standards: OSHA 1910 Subpart D (Walking-Working Surfaces), ANSI MH30.1 (Safety Requirements for the Design, Testing, and Utilization of Portable Dock Leveling Devices)(L2).		
UNIT-V	FIRE SAFETY IN WAREHOUSE.	9
Fire hazards in warehouses (L1) – Fire prevention strategies: housekeeping, storage practices, and electrical safety (L2) – Fire detection and alarm systems (L2) – Fire suppression systems: sprinklers, extinguishers, and fire doors (L2) – Emergency evacuation planning (L3) – Training and drills for fire safety (L3) – Compliance with fire safety regulations and standards (L2) – Case studies of warehouse fire incidents and lessons learned (L2) – Relevant standards: NFPA 13 (Standard for the Installation of Sprinkler Systems)(L2), NFPA 25 (Standard for the Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems)(L2), NFPA 30 (Flammable and Combustible Liquids Code)(L2).		
Total:45 Periods		
OPEN ENDED PROBLEMS /QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination		
Course outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Explain the basic concepts and functions of logistics.	L2-Understand
CO2	Identify and implement safe practices for material handling and storage.	L3 - Apply
CO3	Assess and mitigate risks associated with transportation safety.	L5 - Evaluating
CO4	Apply safety protocols in dock and yard operations.	L3- Apply
CO5	Develop and implement fire safety measures for warehouse environments.	L3- Apply

REFERENCE BOOKS:	
1.	James A. Tompkins and Jerry D. Smith, "The Warehouse Management Handbook," Tompkins Press, 1998.
2.	Michael B. Spear, "Warehouse Safety: A Practical Guide to Preventing Warehouse Incidents and Injuries," CRC Press, 2014.
3.	Kenneth L. Arnold, "Introduction to Materials Handling," Prentice Hall, 1998.
4.	Ministry of Commerce & Industry, "e-Handbook on Warehousing Standards: PM GatiShakti," Government of India, 2022.
5.	Ministry of Commerce & Industry, "Warehousing Development and Regulation Act," Government of India, 2007.
6.	Bureau of Indian Standards (BIS), "IS 16145:2014 - Guidelines for Safety in Warehousing," BIS, 2014.
7.	NITI Aayog, "National Master Plan for Multi-Modal Connectivity: PM GatiShakti," Government of India, 2021.
8.	Daniel E. Della-Giustina, "Fire Safety Management Handbook," CRC Press, 2014.
VIDEO REFERENCES:	
1.	https://www.youtube.com/watch?v=PmR2SKeY9Ms&list=PLGit8yny_3ANzZMsJJjeuxMg-S0f0hGcn
2.	https://www.youtube.com/watch?v=-XRu7BSouvY
WEB REFERENCES:	
1.	https://www.osha.gov/warehousing
2.	https://www.india.gov.in/spotlight/pm-gati-shakti-national-master-plan-multi-modal-connectivity
ONLINE COURSES:	
1.	https://onlinecourses.swayam2.ac.in/ntr24_ed21/preview
2.	https://onlinecourses.nptel.ac.in/noc20_ce09/preview

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1		1	2	1	2	
CO2		2	2	2	2	1
CO3		1	3	1	3	1
CO4	2	1	2	2	2	
CO5		1	3		3	1
Average	2	1.2	2.4	1.5	2.4	1
1-Low, 2-Medium, 3-High.						

ME23IS911	POLYMERS AND COMPOSITE MATERIALS	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	M.E-INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To Impart knowledge on various polymer processing techniques					
2	To Learn about various fibre, Matrix materials and their properties					
3	To Learn the methods by which Polymer matrix composites are made					
4	To Study about the composites used for High temperature applications					
5	To Study the behavior of reinforcements in MMC and PMC					
UNIT-I		PROCESSING OF POLYMERS				6+3
Classification of Polymers (L2) – Properties of Thermo plastics (L2) – Properties of Thermosetting Plastics (L2) – Extrusion (L2) – Injection Moulding(L2) – Blow Moulding(L2) – Compression and Transfer Moulding (L2) – Casting (L2) – Thermo Forming (L2). General Machining properties of Plastics (L2) – Machining Parameters and their effect (L2) – Joining of Plastics (L2) – Thermal bonding (L2) – Applications (L2).						
UNIT-II		FIBERS AND MATRIX MATERIALS				6+3
Fibers – Fabrication, Structure, properties and applications (L2) – Glass fiber (L2), Boron fiber (L2), Carbon Fiber (L2), Organic Fiber (L2), Ceramic and Metallic Fibres (L2) - Whiskers (L2) – Fabrication of Matrix Materials (L2) – Polymers, Metals and Ceramics and their properties (L2) – Interfaces (L2) – Wettability (L2) – Types of Bonding at the Interface (L2) – Tests for Measuring Interfacial Strength (L2) - Physical and Chemical properties (L2).						
UNIT-III		PROCESSING OF POLYMER MATRIX COMPOSITES				6+3
Thermoset Matrix Composites: Hand Layup (L2), Spray (L2), Filament Winding (L2), Pultrusion (L2), Resin Transfer Moulding (L2), Autoclave Moulding(L2) - Bag Moulding (L2), Compression Moulding with Bulk Moulding Compound and SHEET Moulding Compound (L2) – Thermoplastic Matrix Composites (L2) – Film Stacking (L2), Diaphragm Forming (L2), Thermoplastic Tape Laying (L2), Injection Moulding(L2) – Interfaces in PMCs (L2) - Structure, Properties and Application of PMCs (L2) – Recycling of PMCs (L3).						
UNIT-IV		PROCESSING OF METAL MATRIX COMPOSITES				6+3
Metallic Matrices: Aluminium (L2), Titanium (L2), Magnesium (L2), Copper Alloys (L2) – Processing of MMCs: Liquid State (L2), Solid State (L2), In Situ Fabrication Techniques (L2) – Diffusion Bonding (L2) – Powder Metallurgy Techniques (L2) - Interfaces in MMCs (L2) – Mechanical Properties (L2) – Machining of MMCs (L2) – Applications (L2).						

UNIT-V	PROCESSING OF CERAMIC MATRIX COMPOSITES AND CARBON - CARBON COMPOSITES	6+3
Processing of CMCs: Cold Pressing (L2), Sintering (L2), Reaction Bonding (L2), Liquid Infiltration (L2), Lanxide Process (L2) – In Situ Chemical Reaction Techniques: Chemical Vapour Deposition (L2), Chemical Vapour Impregnation (L2), Sol-Gel (L2) – Interfaces In CMCs(L2) – Mechanical Properties and Applications Of CMCs(L2) – Carbon-Carbon Composites (L2) – Applications (L2).		
		Total:45PERIODS
OPEN ENDED PROBLEMS/QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.		
Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Understand the various polymer processing techniques.	L2-Understand
CO2	Understand about various fibre, matrix materials and their properties.	L2-Understand
CO3	Apply the methods by which polymer matrix composites are made.	L3-Apply
CO4	Analyze about the composites used for High temperature applications.	L4-Analyze
CO5	Understand the behavior of reinforcements in MMC and PMC.	L2-Understand
REFERENCE BOOKS:		
1.	Harold Belofsky ,Plastics, Product Design and Process Engineering, Hanser Publishers,2002.	
2.	Mallick,P.K. and Newman.S., Composite Materials Technology, Hanser Publishers,2003.	
3.	Seamour, E.B.Modern Plastics Technology, Prentice Hall,2002.	
4.	M.Balasubramanian , Composite Material and Processing, CRC Press,2014.	
5.	K.K.Chawla, Composite Material: Science and Engineering, Springer, 3 rd Edition,2012.	
VIDEO REFERENCES:		
1.	https://www.youtube.com/watch?v=RihoVfzEfWI	
2.	https://www.youtube.com/watch?v=RMzGBRL_o3E&list=PLSGws_74K01_G67ptndBraskY3jCW7FLQ	
WEB REFERENCES:		
1.	https://www.researchgate.net/publication/319400527_Recent_Advances_in_Metal_Matrix_Composites_MMCs_A_Review	
ONLINE COURSES:		
1.	https://onlinecourses.nptel.ac.in/noc20_me29/preview	

ME23IS913	ADVANCED INTERNAL COMBUSTION ENGINES	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	M.E-INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To gain insight on the working principle of spark ignition engines.					
2	To gain insight on the working principle of compression ignition engines.					
3	To study the pollutant formation and its control in IC engines.					
4	To Study the alternate fuels in IC Engines.					
5	To study the recent technologies adopted in IC engine applications.					
UNIT-I	SPARK IGNITION ENGINES	6+3				
Spark ignition Engine mixture requirements (L2) – Fuel – Injection systems (L2) – Monopoint (L2), Multipoint injection (L2), Direct injection (L2) – Stages of combustion (L2) – Normal and abnormal combustion (L2) – factors affecting knock (L2) – Combustion chambers (L2).						
UNIT-II	COMPRESSION IGNITION ENGINES	6+3				
States of combustion in C.I. Engine (L2) – Direct and indirect injection systems (L2) – Combustion chambers (L2) – Fuel spray behaviour (L2) – spray structure (L2), spray penetration and evaporation (L2) – air motion (L2) – Introduction to Turbo charging (L2).						
UNIT- III	POLLUTANT FORMATION AND CONTROL	6+3				
Pollutant (L2) – Sources (L2) – Formation of carbon monoxide, Unburnt hydrocarbon, NOx, Smoke and Particulate matter (L2) – Methods of controlling Emissions (L2) – Catalytic converters and Particulate Traps (L2) – Methods of measurements and Introduction to emission norms and Driving cycles (L2).						
UNIT-IV	ALTERNATIVE FUELS	6+3				
Alcohol, Hydrogen, Natural Gas and Liquefied Petroleum Gas (L2)- Properties, Suitability, Merits and Demerits as fuels (L2), Engine Modifications (L2).						
UNIT-V	RECENT TRENDS	6+3				
Lean Burn Engines (L2) – Stratified charge Engines (L2) – homogeneous charge compression ignition engines (L2) – Plasma Ignition (L2) – Measurement techniques (L2) – laser Doppler (L2), Anemometry (L2). Use of Nano technology in IC Engines (L2).						
Total:45 PERIODS						

OPEN ENDED PROBLEMS / QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such Problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.		
Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Understand the working principle of spark ignition engine, fuel injections and combustion chambers.	L2-Understand
CO2	Understand the working principle of compression ignition engines, direct and indirect injection systems and turbo charging.	L2-Understand
CO3	Summarize the pollution formation and its control.	L2-Understand
CO4	Understand the alternate fuels in IC Engines.	L2-Understand
CO5	Apply the knowledge about recent technologies adopted in IC engine applications.	L3-Apply
REFERENCE BOOKS:		
1.	Duffy Smith, Auto fuel Systems, The Good Heart Willox Company, Inc., 1989	
2.	Heywood, J.B., Internal Combustion Engine Fundamentals, McGraw-Hill, 1988.	
3.	K.K. Ramalingam, Internal Combustion Engine fundamentals, Scitech Publications, 2002	
4.	R.B. Mathur and R. P. Sharma, Internal Combustion Engines, Dhanapat Rai Publications, 1993.	
VIDEO REFERENCES:		
1.	https://www.youtube.com/watch?v=xTtiBmguhFQ	
WEB REFERENCES:		
1.	https://www.routledge.com/Advances-in-Combustion-Technology/Mishra/p/book/9780367501525	
ONLINE COURSES:		
1.	https://nptel.ac.in/courses/112104033	

ME23IS915	ENGINE POLLUTION AND CONTROL	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	M.E-INDUSTRIAL SAFETY ENGINEERING	Version:1.0				
Course Objectives:						
1	To provide an insight about effect of engine out emissions on human health and environment.					
2	To impart the knowledge on various pollutant species formations in SI and CI engine.					
3	To divulge about various emission measurement techniques in engines and its significance.					
4	To provide a discernment about various emission control methods.					
5	To impart the knowledge about international and national driving cycles and emission standards.					
UNIT-I	AIR POLLUTION-ENGINES	6+3				
Atmospheric pollution from automotive (L2), stationary engines and gas turbines (L2), Global warming (L2) – Greenhouse effect (L2), Effects of engine pollution on human health and environment (L2).						
UNIT-II	POLLUTANT FORMATION	6+3				
Formation of Oxides of nitrogen, Carbon monoxide, Hydrocarbon, Aldehydes, Smoke and Particulate matter emissions (L2). Effects of Engine design and operating variables on emission formation (L2), Noise pollution (L2).						
UNIT-III	EMISSION MEASUREMENT TECHNIQUES	6+3				
CO, CO2 (L2) - Non dispersive infrared gas analyzer (L2), NOx(L2) - Chemiluminescent analyzer (L2), HC (L2) - Flame ionization detector (L2), Smoke (L2) – Opacity and filter paper measurements (L2), Particulate Matter (L2) – Full flow and Partial flow dilution tunnel (L2), Gas chromatography (L2), Noise measurement (L2).						
UNIT-IV	EMISSION CONTROL TECHNIQUES	6+3				
Engine design modifications (L2), Fuel modification (L2), Evaporative emission control (L2), EGR (L2), Air injection (L2), Thermal reactors (L2), Water injection (L2), Common rail direct injection (L2) and Gasoline direct injection system (L2), After. treatment systems (L2) - Catalytic converters (L2), Diesel oxidation catalyst (L2), Particulate traps (L2), De-NOx catalysts (L2), SCR systems (L2). Low temperature combustion concepts (L2).						

UNIT-V	DRIVING CYCLES AND EMISSION STANDARDS	6+3
Transient dynamometer (L2), Test cells (L2), Driving cycles for emission measurement (L2), chassis dynamometer (L2), CVS system (L2), National and International emission standards (L2).		
Total:45PERIODS		
OPEN ENDED PROBLEMS/QUESTIONS		
Course specific open ended problems will be solved during the classroom teaching. Such Problems can be given as assignments and evaluated as internal assessment only and not for the end semester examination.		
Total:45 PERIODS		
Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Understand about atmospheric pollution from engines and its impact on human health and environment.	L2-Understand
CO2	Understand the formation of emissions in both SI and CI engines.	L2-Understand
CO3	Understand the various measurement techniques used globally for the measurement of automotive and stationary.	L2-Understand
CO4	Apply the various control methods / techniques used in IC engine to control the engine out emissions.	L3- Apply
CO5	Apply the transient and steady state driving cycles performed on automotive and stationary engines and emission standards that are followed in the national and international level.	L3- Apply
REFERENCE BOOKS:		
1.	GanesanV., "Internal Combustion Engines", V Edition, Tata McGraw Hill, 2012.	
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3.	George Springer and Donald JPatterson, Engine emissions, Pollutant Formation and Measurement, Plenumpress, 2013.	
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